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RESEARCH ARTICLE

Determination of total dietary fibre and available carbohydrates: A rapid integrated procedure that simulates in vivo digestion

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The new definition of dietary fibre introduced by Codex Alimentarius in 2008 includes resistant starch and the option to include non-digestible oligosaccharides. Implementation of this definition required new methodology. An integrated total dietary fibre method was evaluated and accepted by AOAC International and AACC International (AOAC Methods 2009.01 and 2011.25; AACC Method 32–45.01 and 32–50.01, and recently adopted by Codex Alimentarius as a Type I Method. However, in application of the method to a diverse range of food samples and particularly food ingredients, some limitations have been identified. One of the ongoing criticisms of this method was that the time of incubation with pancreatic α-amylase/amyloglucosidase mixture was 16 h, whereas the time for food to transit through the human small intestine was likely to be approximately 4 h. In the current work, we use an incubation time of 4 h, and have evaluated incubation conditions that yield resistant starch and dietary values in line with ileostomy results within this time frame. Problems associated with production, hydrolysis and chromatography of various oligosaccharides have been addressed resulting in a more rapid procedure that is directly applicable to all foods and food ingredients currently available.

Keywords:
Available carbohydrates / Codex Alimentarius / Dietary fibre determination / Enzymic / Non-digestible oligosaccharides / Resistant starch

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1 Introduction

Interest in dietary fibre is a consequence of the belief that dietary fibre contributes positively to the health/quality of life of the consumer. Since the definition and analysis of dietary fibre are intimately related, analysis methods should be developed to comply with the conceptual definitions. However, compromises must be accepted due to constraints of cost and time [1].

Available carbohydrates are carbohydrates that are readily hydrolysed into D-glucose and D-fructose and absorbed into the human small intestine. They consist of non-resistant starch, maltodextrins, sucrose, the glucose component of lactose and free D-glucose and D-fructose.

The term 'dietary fibre' was coined by Hipsley [2] to cover the non-digestible constituents of plants that make up the plant cell wall, known to include cellulose, hemicellulose and lignin. The aim was to define some property of the constituent of the food that could be related to physiological behaviour in the human small intestine. This definition of dietary fibre was broadened by Trowell et al. [3] to become primarily a physiological definition, based on edibility and resistance to digestion in the human small intestine; the definition included indigestible polysaccharides such as

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Abbreviations: ACH, available carbohydrates (= non-resistant starch + maltodextrins + maltose + sucrose + free D-glucose and D-fructose + the D-glucose component of lactose); **DF,** dietary fibre (= HMWDF + LMWDF) (= IDF + HMWSDF + LMWSDF); **HMWDF,** high molecular weight dietary fibre (= IDF + HMWSDF); **HMWSDF,** high molecular weight soluble dietary fibre (= SDFP); **IDF,** insoluble dietary fibre; **LMWSDF,** low molecular weight soluble dietary fibre (= SDFS); **NDO,** non-digestible oligosaccharides (= LMWSDF = SDFS); **RS,** resistant starch; **SDFP,** soluble dietary fibre which precipitates in the presence of 76% ethanol; **SDFS,** soluble dietary fibre that remains soluble in the presence of 76% ethanol

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gums, modified celluloses, mucilages and pectin, and non-digestible oligosaccharides (NDO).

Initial efforts to develop a method to satisfy this analytical requirement focused on the removal of starch and protein. Enzymes employed had to meet specific activity requirements and be devoid of contaminating enzymes active on dietary fibre components. The method that evolved was AOAC Official Method 985.29 'Total Dietary Fiber in Foods; Enzymatic-Gravimetric Method' [4, 5]. Subsequently, the method was extended to allow measurement of total, soluble and insoluble dietary fibre in foods (AOAC Official Method 991.43) [6], and various other modified methods for fibre determination have been approved by AOAC International [7] and accepted by Codex Alimentarius [8, 9].

In parallel research in the UK, Englyst et al. [10] developed methods for the measurement of available starch and non-starch polysaccharides (NSP) [11–14], based on original work of Southgate [15, 16]. This NSP procedure measures only non-starch polysaccharides; resistant starch (RS) is deliberately removed and NDOs are not measured.

Based on the recommendation for endorsement of the Codex Committee on Nutrition and Foods for Special Dietary Uses (CCNFSDU) in November 2008, a definition for dietary fibre was adopted in June 2009 by the Codex Alimentarius Commission (CAC) [17]. The definition includes carbohydrate polymers that are not hydrolysed by the endogenous enzymes in the small intestine of humans and thus includes RS. However, decisions concerning the inclusion, or not, of oligosaccharides of degrees of polymerisation (DP) of 3–9 were left to the discretion of national authorities [18].

A method that appeared to satisfy this definition was published in 2007 [19] and this method was successfully evaluated in interlaboratory evaluations [20, 21]. In this method, samples are incubated with pancreatic α-amylase (PAA) and amyloglucosidase (AMG) under near physiological conditions (37°C, pH 6) as previously described for measurement of resistant starch (AOAC Method 2002.01) [7, 22, 23]. This method allows the measurement of high molecular weight dietary fibre (HMWDF) which included insoluble dietary fibre (IDF) and high molecular weight soluble dietary fibre (HMWSDF) which precipitates in the presence of 78% aqueous ethanol (SDFP) and also low molecular weight soluble dietary fibre (LMWSDF) that remains soluble in the presence of 78% aqueous ethanol (SDFS). However, in the application of this method to a range of food products and ingredients, several challenges/concerns were identified. Firstly, an incubation time with PAA plus AMG of 16 h was considered not to simulate physiological conditions. Based on numerous studies in literature, a more likely residence time for food in the small intestine is 4 ± 1 h [24–33]. Secondly, most commercially available fructo-oligosaccharides (FOS) contain the trisaccharide, fructosyl-β-(2-1)-fructosyl-β-(2-1)-fructose (F3). This oligosaccharide elutes from the Waters Sugar-Pak® (Waters Ireland, Woodford Business Park, Santry, Dublin, Ireland) column at the same point as disaccharides such as lactose, maltose and sucrose, so it is not included in the analytical value for SDFS (DP > 3). A solution to this problem is to use different chromatographic columns as are described in AOAC Method 2001.03 [7, 34]. Thirdly, on hydrolysis of products high in starch, including certain breads and rice products, various maltodextrins are produced which are very resistant to further hydrolysis by AMG or PAA [35–37]. However, these oligosaccharides are readily hydrolysed [8] by a mucosal α-glucosidase preparation from the small intestine of pig [38], indicating that they should not be included in the SDFS fraction of dietary fibre (DF). A fourth challenge experienced in the use of AOAC Method 2009.01 is the considered underestimation of phosphate cross-linked starch (Resistant Starch 4, *e.g.* FiberRite® and Fibersym®, MGP Ingredients, Cray Business Park, Atchison, Kansas, USA). Much higher DF values for FiberRite® and Fibersym® are obtained [39] using the Prosky TDF method (AOAC Method 985.29) [5]. A fifth concern in the use of AOAC Method 2009.01, relates to the use of sodium azide as a preservative in the buffer. While the concentration employed is low (0.02% w/v), it is still of concern to some analysts. With the extended incubation conditions described in AOAC Method 2009.01 (16 h, 37°C, pH 6), inclusion of an antimicrobial agent is essential. However, with a shorter incubation time of just 4 h, the sodium azide could potentially be removed altogether or alternatively replaced with another preservative such as a few drops of toluene.

The aim of this research was to develop a method for the measurement of dietary fibre that resolves each of the challenges detailed above. Part of this process involved a comparison of AOAC Method 2002.02 [22] and the Englyst et al. [10] methods for measurement of resistant starch, in an attempt to understand why the methods give similar RS values across a range of samples, even though the incubation times with PAA/AMG are very different. The ultimate goal being to develop a method that more closely simulates physiological conditions. Problems associated with measurement of FOS and production of resistant maltodextrins from starch have also been addressed. A second aim was to incorporate quantitative measurement of available carbohydrates into the overall analysis.

2 Materials and methods

2.1 Materials

D/L-Maleic acid (cat. no. M-0375), bovine serum albumin (cat. no. A-2153), dimethyl sulphoxide (cat. no. D-8779) and sodium azide (cat. no. S-8032) were from Sigma–Aldrich Ireland Ltd. (Dublin, Ireland). Acetic acid (glacial) GR, sodium hydroxide and calcium chloride ($CaCl_2.2H_2O$) were

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from Merck (Darmstradt, Germany). Partially degraded chicory inulin (Raftilose P-95®) was a kind gift from Raffinerie Tirlemontoise S.A. (Tienen, Belgium). Galacto-oligosaccharides (Purimune high purity GOS) were obtained from GTC Nutrition, CO. Polydextrose® (Litesse®) was from Danisco (Copenhagen, Denmark) and Fibersol 2® and DE-25 corn syrup (Pinedex®) were from Matsutani Chemical Company, Hyogo, Japan. AdvantaFiber® (a mixture of isomalto-oligosaccharides) was from Top Health Ingredients, Inc., Edmonton, Alberta, Canada. Coupling Sugar® (a mixture of maltotriosylsucrose, maltosylsucrose, glucosylsucrose, maltotriose and sucrose) was from Hayashibara Co. Ltd., Okayama Japan Regular Maize Starch (Lot 60401; RMS) was from Penford Australasia, Lane Cove, NSW, Australia. Hylon VII® (Ref. 98GH8401), Novelose 330® (Ref. AH17529) and Novelose 240® (Ref. 96LF10063) were from National Starch and Chemical Company, Bridgewater, CT. Native potato starch was from Avebe, Foxhol, The Netherlands. ActiStar® (enzyme-modified tapioca/cassava starch; US Patent 6,043,229) was from Cerestar, Vilvoorde, Belgium. Potato amylose (cat. no. A-9262) and ACS Soluble starch (cat. no. S-9765) were from Sigma Chemical Company (St. Louis, MO, USA). Amyloglucosidase (AMG) (cat. no. E-AMGFR), pancreatic α-amylase (PAA: Megazyme cat. no. E-PANAA), PAA/AMG mixture (PAA 60 KU/g plus AMG 17 KU/g; Megazyme cat. no. E-PAAMG), thermostable α-amylase (cat. no. E-BLAAM), protease (cat. no. E-BSPRT), barley β-glucan (medium viscosity; cat. no. P-BGBM), citrus pectin (cat. no. P-CITPN), wheat arabinoxylan (cat. no. P-WAXM), D-fructose/D-glucose assay kit (cat. no. K-FRUGL), α-Amylase assay kit (Ceralpha®; cat. no. K-CERA), Total Starch assay kit (cat. no. K-TSTA), TDF assay kit (cat. no. K-TDFR), glycerol assay kit (cat. no. K-GCROLGK), D-Sorbitol assay kit (cat. no. K-SORB), Resistant Starch assay kit (cat. no. K-RSTAR), Amberlite FPA53 (OH^-)(cat. no. G-AMBOH) and Amberlite 200C (H^+)(cat. no. G-AMBH) were obtained from Megazyme International Ireland Limited, Bray, Ireland). Uncle Ben's Ready Rice, white, extra and RTH white rice were obtained from Professor William Park, Texas A & M University, College Station, TX. Long grain rice rice, cream flour (wheat), Ryvita® crackers, oat bran, Weetabix®, Kellogg® All Bran®, Quick oats®, Sugar Frosties®, Kellogg® corn flakes, whole wheat pasta, tinned butter beans, sweet corn, garden peas, red kidney beans, Heinz® baked beans, and chick peas, fresh cabbage, broccoli, semi-ripe banana, carrots and potato were obtained from a local supermarket. Potato was cooked in boiling water for 30 min, mashed and freeze dried. All fresh vegetables were sliced into thin sections, freeze dried, milled to pass a 0.5 mm screen and stored in Duran® air-tight bottles at room temperature. Canned beans and vegetables were poured onto a strainer and washed with demineralised water, freeze-dried and milled to pass a 0.5 mm screen. Dry breakfast cereals were milled to pass a 0.5 mm screen and stored in air-tight Duran® bottles.

2.2 Methods

2.2.1 Assay of PAA and AMG and definitions of enzyme activity

PAA was measured using the Ceralpha® assay procedure employing benzylidene blocked *p*-nitrophenyl maltoheptaoside in the presence of excess levels of thermostable α-glucosidase. Incubations were performed in sodium maleate buffer at pH 6.9 and 40°C as described in the Ceralpha® kit booklet (cat. no. K-CERA; AOAC Official Method 2002.01). One Unit of enzyme activity (CU) is defined as the amount of enzyme that releases 1 μmol of *p*-nitrophenol/min under the defined assay procedure. A British Pharmacopeia Units (BPU) of α-amylase is defined as that amount of enzyme that 'decomposes starch at an initial rate such that one micro-equivalent of glycosidic linkages is hydrolysed per minute'. For pancreatic α-amylase, one CU = 2.7 BPU. AMG was assayed by incubating 0.2 mL of suitably diluted enzyme in 100 mM sodium acetate buffer (pH 4.5) with 0.5 mL of soluble starch (10 mg/mL) in 100 mM sodium acetate buffer (pH 4.5) at 40°C. At various time intervals, reaction tubes were heated to ~100°C in a boiling water bath to terminate the reaction and released glucose was measured using GOPOD reagent (Glucose assay kit; Megazyme cat. no. K-GLUC). One unit of AMG is defined as the amount of enzyme required to release 1 μmol of glucose per minute from starch (10 mg/mL) at pH 4.5 and 40°C. When in admixture with PAA, AMG was assayed using AMG Assay Reagent (Megazyme cat. no. R-AMGR3, employing *p*-nitrophenyl-β-maltoside as substrate) and the units of activity on starch were calculated using a conversion factor. One unit of activity on *p*-NP-β-maltoside = 11.5 units of activity on starch.

2.2.2 Hydrolysis of starch containing samples with PAA and AMG and measurement of non-resistant starch

2.2.2.1 Incubation in polypropylene centrifuge tubes

Samples (1.00 g) of pure starches or starch containing materials were accurately weighed into 50 mL polypropylene, screw cap, centrifuge tubes (SciQuip, Shropshire, UK, 50 mL popypropylene tube 106.7 mm × 28.8 mm, Cat. No. SQ-3119-0050; as described by Englyst et al. [10]). To each tube, guar gums (50 mg, Sigma Chemical Co. Cat. No. G-4129), together with five glass marbles, were added. Sodium acetate buffer (20 mL, 100 mM, pH 5.2) was added, the tubes capped and the contents shaken by hand and pre-incubated for 10 min at 37°C. Enzyme preparation (5 mL, in sodium acetate buffer, pH 5.2) was added and the tubes were tightly capped and placed into a shaking water bath and incubated at 37°C with linear shaking at 150 rpm. At various time intervals after addition of the enzyme mixture, the tubes were removed from the bath and samples (0.5 mL) of the

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well-suspended reaction mixture were removed with a positive displacement dispenser and added to 20 mL of 66% v/v aqueous ethanol and mixed thoroughly. Samples of these suspensions were centrifuged at 1500*g* and aliquots (0.1 mL) of the supernatant solutions were analysed for glucose using GOPOD Reagent. Data were analysed using an Excel calculator (Supporting Information Fig. S1).

Variations on these incubations included the following:

(1) Incubations were performed with and without including the guar gum and marbles in the reaction tubes.
(2) Replacement of sodium acetate buffer (100 mM, pH 5.2) with sodium maleate buffer (100 mM, pH 6.0) plus BSA (0.5 mg/mL) and sodium azide (0.02% w/v) *[2.2.4.1.3(10)]*.
(3) Incubations were performed with varying amounts of PAA (1.5–24 KU/incubation) or AMG (0.34–3.4 KU/incubation).

2.2.2.2 Incubation in Fisherbrand® glass bottles

Samples (1.00 g) of pure starch or starch containing materials were accurately weighed into 250 mL Fisherbrand® soda glass, wide-mouthed bottles. The sample was wet with 1 mL of ethanol (95% v/v) and 35 mL of either sodium acetate buffer (100 mM, pH 5.2) or sodium maleate buffer (100 mM, pH 6.0) *[2.2.4.1.3(10)]* was added and the bottles placed into a water bath at 37°C for 10 min to equilibrate to temperature. Bottle contents were either stirred at 170 rpm on a 2mag Mixdrive 15® (2mag AG, Munich, Germany) [with a $7 \times 30\,mm^2$ magnetic stirrer bar added directly to the bottle, or with a magnetic stirrer bar suspended from the bottle cap (to allow stirring without grinding of the sample between the stirrer bar and the bottom of the bottle)] (Fig. 1), or were shaken at 150 rpm in orbital motion in a Grant OLS 200® (Grant Instruments, Cambridge, UK) shaking water bath. Enzyme preparation {5 mL, in 100 mM sodium acetate buffer, pH 5.2 or 100 mM sodium maleate buffer, pH 6.0 *[2.2.4.1.3(10)]*} was added, the bottles were tightly capped and placed back into the water bath and either shaken or stirred at 37°C as above. The enzyme preparation (5 mL) contained PAA (1.7–14 KU) and AMG (0.34–3.4 KU). At various time intervals after the addition of the enzyme mixture, the bottles were removed from the bath one at a time and samples (0.5 mL) of the well suspended reaction mixture were removed with a positive displacement dispenser and added to 10 mL of 66% v/v aqueous ethanol and mixed well. Samples of these suspensions were centrifuged at 1500*g* and aliquots (0.1 mL) of the supernatant solutions were analysed for glucose using GOPOD Reagent. Data were analysed using an Excel calculator (Supporting Information Fig. S2).

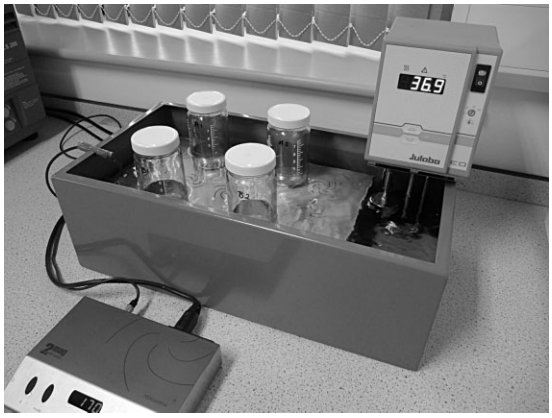

Figure 1. Picture showing a 2mag Mixdrive 15® submersible magnetic stirrer in a custom made water bath. This allows stirring of 15 samples at controlled speed and 37°C.

2.2.3 Hydrolysis of starch containing samples with PAA and AMG and measurement of resistant starch

Samples were either analysed as received, or alternatively they were pre-cooked in buffer before analysis. In the latter case, samples (1.00 g) were accurately weighed into Fisherbrand® soda glass, wide-mouthed bottles *[2.2.4.1.2(2)]* and wet with 1 mL of ethanol to aid subsequent dispersion. Sodium maleate buffer (35 mL, pH 6.0) was added and the samples were suspended by swirling. The bottles were loosely capped and incubated in a boiling water bath for 15 min with occasional gentle swirling by hand. Bottles were removed from the boiling water bath and placed in a water bath set at 37°C to equilibrate over 10 min, and then enzymes were added and incubations were performed according to the standard procedure described above *[2.2.2.2]*. After 4 h, an aliquot (4 mL) of the hydrolysate was removed with a positive displacement dispenser from the stirring suspension. This was added to 4 mL of ethanol (or IMS) and stirred vigorously. The suspension was centrifuged and RS recovered and washed according to AOAC Method 2002.02 [22, 23]. The RS was dissolved in KOH, neutralised, hydrolysed to glucose and the glucose measured with GOPOD Reagent.

2.2.4 Measurement of total dietary fibre (including resistant starch) and available carbohydrates in cereal, vegetable, fruit and food products (RINTDF procedure)

The rapid integrated TDF method described here (RINTDF) is based on an integrated method developed by McCleary [19] but both the incubation time and concentration of both PAA and AMG have been altered to overcome a few limitations found with that protocol. The HPLC was routinely performed with Toyo TSK-GELR G2500PWXL® (Tosoh Corporation) gel permeation column (in place of a Waters Sugar-Pak® column) to improve separation of FOS components. An alternative, more rapid procedure for deionisation of samples for HPLC is proposed. The procedure measures HMWDF, that includes IDF and HMWSDF (SDFP) and

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SDFS (non-digestible oligosaccharides—NDO). The method can simply be adapted to separately measure IDF and SDFP if required. This method update essentially resolves all of the analytical challenges experienced with AOAC Methods 2009.01 and 2011.25 and greatly simplifies measurement of SDFS.

2.2.4.1 Sample treatment and determination of HMWDF

2.2.4.1.1 Principle

Food samples should be analysed 'as eaten'. Wet or moist samples should be freeze dried and milled to pass a 1 mm screen; pasta should be cooked, freeze dried and milled; bread samples should be freeze dried and milled; etc. 'Wet' samples are not used in the analytical method because it is essential to have an accurate measurement of the original sample weight. Freeze drying is considered to be the most gently drying method available. Sample (1 g) is weighed into a 250 mL Fisherbrand® bottle and sodium maleate buffer (pH 6) containing PAA and AMG is added and the contents stirred at 170 rpm on a submersible 2mag Mixdrive 15® magnetic stirrer, or are shaken in a shaking incubation bath at 150 rpm in orbital mode, for exactly 4 h at 37°C. During this time, non-resistant starch is solubilised and hydrolysed to D-glucose by the combined action of the two enzymes. The pH is adjusted to approximately 8 to minimise the action of AMG and then the bottles are incubated at 100°C to completely inactivate PAA and AMG and to denature protein. Denatured protein is hydrolysed with protease at 60°C and then pH is adjusted to approximately 4.5 with acetic acid. [If available carbohydrates (ACH) is to be determined, a 0.5 mL sample is removed at this point]. Four volumes of ethanol are added with stirring, to precipitate soluble, polymeric dietary fibre (SDFP) and resistant starch (that is solubilised, but not depolymerised, in the 100°C incubation step) and to remove depolymerised protein and D-glucose (derived from depolymerised, non-resistant starch). The suspension is filtered and the residue on the filter is washed with 76% ethanol, 95% ethanol and acetone; dried, and weighed. One duplicate is analysed for protein and the other for ash. HMWDF is the weight of the filtered and dried residue less the weight of the protein and ash. The ethanolic wash solutions are concentrated, desalted and analysed by HPLC to determine SDFS content.

2.2.4.1.2 Apparatus

(1) Grinding mill.—Centrifugal, with 12-tooth rotor and 0.5 mm sieve, or similar device. Alternatively, a cyclone mill can be used for small test laboratory samples provided the mill has sufficient air flow or other cooling to avoid overheating of samples.

(2) Digestion bottles.—250 mL Fisherbrand® soda glass, wide mouth bottles with polyvinyl lined cap (cat. no. 11798859) (https://extranet.fisher.co.uk/insight2_uk/mainSearch.do?keywords=FB73219&utm_source=fisher_web&utm_medium=product_page&utm_campaign=all_product_promote) (accessed 2 December 2014).

(3) Fritted crucible.—Gooch, fritted disc, Pyrex® (Corning Incorporated Life Sciences, Tewksbury, MA, USA) 50 mL, pore size coarse, ASTM 40–60 mm, Corning® No. 32940-50C, or equivalent.

(4) Prepare as follows:

   (a) Ash overnight at 525°C in muffle furnace, cool furnace to 130°C before removing crucibles to minimise breakage.
   (b) Remove any residual Celite® and ash material by using a vacuum.
   (c) Soak in 2% cleaning solution *[2.2.4.1.3(15)]* at room temperature for 1 h.
   (d) Rinse crucibles with water and deionised water.
   (e) For final rinse, use 15 mL acetone and air dry.
   (f) Add approximately 1.0 g Celite® to dried crucibles and dry at 130°C to constant weight.
   (g) Cool crucible in desiccator for approximately 1 h and record mass of crucible containing Celite®.

(5) Filtering flask.—Heavy-walled, 1 L Büchner flask (Fig. 2).

(6) Rubber ring adaptors.—For use to join crucibles with filtering flasks (Fig. 2).

(7) Vacuum source.—Vacuum pump or aspirator with regulator capable of regulating vacuum (*e.g.* Edwards XDS 10, Edwards Vacuum Engineering, Crawley, West Sussex, UK); single-phase 115/230V; product code: A726-01-903).

(8) Water bath(s).—A 2mag Mixdrive 15® submersible magnetic stirrer with a $30 \times 7\,mm^2$ stirrer bar, set at 170 rpm (Fig. 1) or alternatively a rotary motion

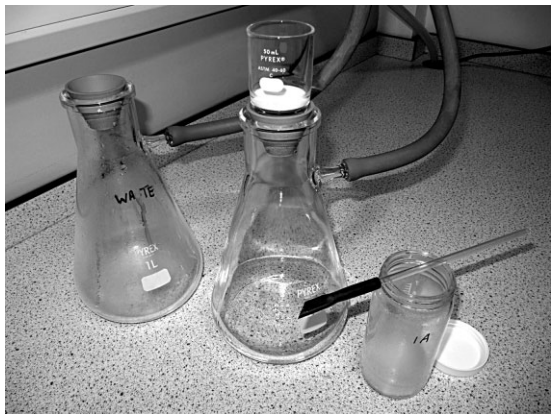

Figure 2. Buchner flask showing Gooch®, fritted disc funnel and rubber policeman spatula.

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(150 rpm), large-capacity (20–24 L) water bath with cover; capable of maintaining temperature of 37 ± 1°C and 60 ± 1°C (*e.g.* Grant® OLS 200 shaking incubation bath).

(9) Balance.—0.1 mg readability, accuracy and precision.
(10) Ovens.—Two, mechanical convection, set at 103 ± 2° and 130 ± 3°C.
(11) Timer.
(12) Desiccator.—Airtight, with silica gel or equivalent desiccant. Desiccant dried biweekly overnight in 130°C oven.
(13) pH meter.
(14) Thermometer.—Capable of measuring to 110°C.
(15) Positive displacement pipettor.—*e.g.* Eppendorf Multipette® (Eppendorf UK Limited, Arlington Business Park, Stevenage, UK) with 25 mL Combitip® (to dispense 5 mL aliquots of PAA/AMG solution, 3 mL aliquots of 0.75 M Tris base solution and 4 mL aliquots of 2 M acetic acid); and with 5.0 mL Combitip® (to dispense 0.1 mL of protease solution).
(16) Cylinder.—Graduated, 100 and 500 mL.
(17) Magnetic stirrers and stirring bars.—(7 × 30 mm^2; plain magnetic stirrer bars; cat. no. 442-0269, VWR Dublin, Ireland).
(18) Rubber policeman spatulas.—VWR International (cat. no. 53801-008) (Fig. 2).
(19) Muffle furnace.—525 ± 5°C.
(20) Polypropylene columns.—Bio-Rad, Econo-Pac™ Disposable Chromatography Columns (cat. no. 732-1010) with an Alltech One-Way Stopcock (cat. no. 211524) (Fig. 3) and a Gilson Minipuls Evolution four channel pump (http://www.gilson.com/en/Pipette/Products/63.229/Default.aspx#.VH2qw97ycRk).
(21) Liquid chromatograph (LC).—With oven to maintain a column temperature of 80°C and a 50 μL injection loop. System must separate maltose from maltotriose.

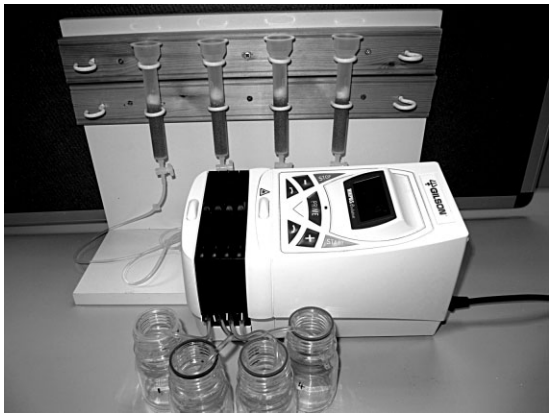

Figure 3. Bio-Rad Eco-Pac™ disposable chromatography columns with Alltech® One-Way Stopcock, containing ion exchange resins with cotton wool wad on top. Connected to a Gilson Minipuls *continuum*® multi-channel pump.

(22) Guard column (or pre-column).—*Option 1:* TSK® guard column PWXL 6.0 mm id × 4 cm (http://www.separations.us.tosohbioscience.com/Products/HPLCColumns/SizeExclusion/WaterSolublePolymers/PWxl/TSKgel+G2500PWxl.htm. Sigma Chemical Co. https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/Supelco/Product_Information_Sheet/8523.pdf).
Option 2: Waters Guard Pak LC pre-column inserts (Part No. WAT015209; Milford, MA).
(23) LC column.—*Option 1*: Two LC columns connected in series. TSK-GELR G2500PWXL®, 7.8 mm id × 30 (http://www.separations.us.tosohbioscience.com/Products/HPLCColumns/SizeExclusion/Water SolublePolymers/PWxl/TSKgel+G2500PWxl.htm Sigma Chemical Co. https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/Supelco/Product_Information_Sheet/8524.pdf) or equivalent; flow rate 0.5 mL/min; column temp. 80°C; run time 30 min to assure column cleaned out.
Option 2: Waters Sugar-Pak®.—6.5 × 300 mm (Part No. WAT085188).
(24) Detector.—Refractive index (RI); maintained at 50°C.
(25) Data integrator or computer.—For peak area measurement.
(26) Filters for disposable syringe.—Millipore Millex® Syringe Driven Filter Unit 0.45 mm (low protein binding Durapore PVDF), 25 or 13 mm or equivalent.
(27) Filters for water.—Millipore, 0.45 mm Durapore® Membrane Filters type HVLP, 47 mm.
(28) Filter apparatus.—To hold 47 mm, 0.45 mm filter; to filter larger volumes of water.
(29) Syringes.—10 mL, disposable, plastic.
(30) Syringes.—Hamilton® 100 μL, 710SNR syringe.
(31) Rotary evaporator.—Heidolph Laborota® 4000 or equivalent.

2.2.4.1.3 Reagents

(1) Ethanol (or industrial methylated spirits; IMS) 95% v/v.
(2) Ethanol (or IMS) 78% v/v.—Place 180 mL deionised water into 1 L volumetric flask. Dilute to volume with 95% v/v ethanol (or IMS). Mix.
(3) Acetone, reagent grade.
(4) PAA plus AMG powder.—PAA (60 KU/g) plus AMG (17 KU/g) as a freeze-dried powder mixture (Megazyme cat. no. E-PAAMG). (Note: 1 U of AMG activity is the amount of enzyme required to release one μmole of D-glucose from soluble starch per minute at 40°C and pH 4.5; 1 U of PAA activity is the amount of enzyme required to release one μmole of *p*-nitrophenyl from Ceralpha reagent per minute at 40°C and pH 6.9). PAA/AMG preparations should be essentially devoid of β-glucanase, β-xylanase and detectable levels of free D-glucose. Stable for >4 years at −20°C.

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(5) PAA (6 KU/5 mL)/AMG (1.7 KU/5 mL).—Immediately before use, dissolve 1 g of PAA/AMG powder in 50 mL of sodium maleate buffer (50 mM, pH 6.0 plus 2 mM $CaCl_2$ and 0.02% sodium azide) and stir for approximately 5 min. Store on ice during use and use within 4 h of preparation.

(6) Protease solution (50 mg/mL).—dissolve 500 mg of protease (Megazyme cat. no. E-BSPRDR; ~ 6 U/mg) in 10 mL of distilled water (~300 Tyrosine Units/mL). Solution is viscous; dispense using a positive displacement dispenser. Protease must be devoid of α-amylase and essentially devoid of β-glucanase and β-xylanase. Use as supplied. Powder is stable for >3 years at −20°C. Liquid should be stored at −20°C and is stable to repeated freezing and thawing.

(7) LC retention time standard.—A mixture of corn syrup solids (DP >3; DE 25; Matsutani Chemical Industry Co., Ltd, Itami City, Hyogo, Japan; www.matsutani.com) and maltose in a ratio of 4:1 (w/w). Dissolve 2.5 g of oligosaccharide mixture in 80 mL of 0.02% sodium azide solution and transfer to 100 mL volumetric flask. Pipette 10 mL of glycerol internal standard into the flask. Bring to volume with 0.02% sodium azide solution. Transfer solutions to 50 mL polypropylene storage bottles. Stable for >1 year at room temperature. Stable for > 4 years at −20°C.

(8) Glycerol internal standard (for TSK® gel permeation column).—100 mg/mL containing sodium azide (0.02% w/v). Weigh 100 g of analytical grade (>99%) glycerol into a 1 L volumetric flask. Dissolve in 800 mL of 0.02% (w/v) sodium azide solution *[2.2.4.1.3(13)]* and adjust to volume with 0.02% sodium azide solution. Mix well. Stable for >2 years at room temperature. Stable for >4 years at −20°C. (Note: Handle sodium azide with caution, only after reviewing MSDS, using appropriate personal protective gear and laboratory hood).

(9) D-Glucose LC standards (5, 10, 20 mg/mL).—Accurately weigh 0.5, 1.0 and 2.0 g portions of high purity (> 99.5%) D-glucose (Sigma Chemical Company; cat. no. 5767) and transfer to three separate 100 mL volumetric flasks. To each flask pipette 10 mL of internal standard *[2.2.4.1.3(8)]*. Bring to volume with 0.02% sodium azide solution. Transfer solutions to 100 mL Duran® bottles. Stable at room temperature for 1 year.

(10) Sodium maleate buffer.—50 mM, pH 6.0 plus 2 mM $CaCl_2$ and 0.02% sodium azide. Dissolve 11.6 g of maleic acid in 1600 mL of deionised water and adjust the pH to 6.0 with 4 M (160 g/L) NaOH solution. Add 0.6 g of calcium chloride ($CaCl_2.2H_2O$) and 0.4 g of sodium azide and adjust the volume to 2 L. Stable for >1 year at 4°C. (Note: Do not add the sodium azide until the pH has been adjusted. Acidification of sodium azide releases a poisonous gas. Handle sodium azide and maleic acid with caution, only after reviewing MSDS, using appropriate personal protective gear and laboratory hood).

(11) Tris base (Megazyme cat. no. B-TRIS500).—Add 90.8 g of Tris base to approximately 800 mL of deionised water and dissolve. Adjust the pH to 11.0 if necessary. Adjust the volume to 1 L. Stable for >1 year at room temperature.

(12) Acetic acid solution, 2 M.—Add 115 mL of glacial acetic acid (Fluka 45731) to a 1 L volumetric flask. Dilute to 1 L with deionised water. Stable for >1 year at room temperature.

(13) Sodium azide solution (0.02% w/v).—Add 0.2 g of sodium azide to 1 L of deionised water and dissolve by stirring. (Note: Do not add sodium azide to solutions of low pH. Acidification of sodium azide releases a poisonous gas. Handle sodium azide with caution, only after reviewing MSDS, using appropriate personal protective gear and laboratory hood.) Stable at room temperature for >2 years.

(14) Deionised water containing $Na_2CaEDTA$ (50 mg/L). (Only required if HPLC is performed using a Waters Sugar-Pak® column).—Weigh 50 mg of $Na_2CaEDTA$ into a 1 L Duran bottle and dissolve in 1 L distilled water. Prepare fresh weekly; filter through 0.45 mm filter before use.

(15) Cleaning solution.—Micro-90® (International Products Corp. (www.ipcol.com/shopexd.asp?id=15) (accessed 4 October 2012). Make a 2% solution with deionised water.

(16) pH standards.—Buffer solutions at pH 4.0, 7.0 and 10.0.

(17) Celite®.—Acid-washed, pre-ashed (Megazyme G-CEL100 or G-CEL500).

(18) Mixed-bed ion exchange resins for each test portion.—

(a) m-1.—Approximately 4 g Amberlite® FPA53 (OH^-) resin (Megazyme cat. no. G-AMBOH), ion exchange capacity 1.6 meq/mL (minimum) and

(b) m-2.—Approximately 4 g Ambersep® 200 (H^+) resin (Megazyme cat. no. G-AMBH), ion exchange capacity: 1.6 meq/mL (minimum). Mix the two resins just prior to use and pack in column *[2.2.4.1.2(20)]*, Bio-Rad disposable chromatography column) for analysis of each test portion (see Fig. 3). After mixing and packing, add a porous frit (as supplied with the columns) and push into place with a spatula. Wash the resins with 20 mL of deionised water. If using other resins and there is a concern that carbohydrates may be retained on the resin, prepare a test solution consisting of 1 mL of 100 mg/mL internal standard *[2.2.4.1.3(8)]*, and 2.5 mL of 10 mg/mL fructo-oligosaccharides

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diluted to 10 mL. Proceed to step *[2.2.4.2.2(3)]*, 'Deionisation of sample'. Recovery of the internal standards and fructo-oligosaccharides should match that of the solution injected directly onto the LC.

Reagents (4), (6), (7) and (8) are available from Megazyme International Ireland in the 'Rapid Integrated TDF Kit (cat. no. K-RINTDF).

2.2.4.1.4 Preparation of test samples

Samples were collected and prepared as intended to be eaten, *i.e.* baking mixes were prepared and baked, and pasta was cooked, etc. Samples were defatted according to AOAC Method 985.29 if > 10% fat. High-moisture samples (>25%) were freeze dried. Samples ca. 50 g were ground in a grinding mill, *[2.2.4.1.2(1)]* to pass a 0.5 mm sieve. All materials were transferred to a wide-mouthed plastic jar, sealed and mixed well by shaking and inverting and then stored in the presence of a desiccant.

2.2.4.1.5 Enzyme purity

To ensure the absence of undesirable enzymatic activities and effectiveness of desirable enzymatic activities, standards listed in AOAC Method 985.29 (namely β-glucan, pectin, larch galactan, wheat starch, high amylose maize starch and casein) were analysed each time an enzyme lot changed or at a maximum of a 6 month interval.

2.2.4.1.6 Enzymatic digestion of sample

Blanks.—With each assay, run two blanks along with samples to measure any contribution from reagents to residue.

Samples.—

(1) Weigh.—*Duplicate samples* 1.000 ± 0.005 g samples accurately into Fisherbrand® glass bottles.
(2) *Addition of enzymes*.—Wet the sample with 1.0 mL ethanol (or IMS) and add 35 mL of 100 mM sodium maleate buffer *[2.2.4.1.3(10)]* and a $7 \times 30\,mm^2$ stirrer bar to each bottle and place these on a 2mag Mixdrive 15 magnetic stirrer apparatus *[2.2.4.1.2(8)]* (Fig. 1) suspended in a water bath set at 37°C. Stir the contents at 170 rpm for 10 min to equilibrate to 37°C. Alternatively, transfer the bottles to a Grant OLS 200 shaking incubation bath (or similar) and secure the bottles in place with the springs in the shaker frame and shake at 150 rpm in orbital motion.
(3) *Incubation with PAA/AMG*.—Add 5 mL of PAA/AMG solution *[2.2.4.1.3(5)]* to each bottle, cap the bottles and incubate the reaction solutions at 37°C with stirring at 170 rpm using a magnetic stirrer bar and a 2mag Mixdrive 15 magnetic stirrer apparatus; alternatively incubate in a shaking water bath maintained at 37°C at 150 rpm (orbital motion) for exactly 4 h.
(4) *Adjustment of pH to approximately 8.2 (pH 7.9–8.4); inactivation of PAA and AMG*.—After exactly 4 h, remove all sample bottles from the stirring or shaking water bath and immediately add 3.0 mL of 0.75 M Tris base (pH 11) *[2.2.4.1.3(11)]* to terminate the reaction. At the same time, if only one shaker/stirrer bath is available, increase the temperature of the incubation bath to 60°C in readiness for the protease incubation step. Slightly loosen the caps of the sample bottles, place the bottles in a water bath (non-shaking) at 95–100°C, and incubate for 20 min with occasional shaking by hand. Using a thermometer, ensure that the final temperature of the bottle contents is >90°C (checking of just one bottle is adequate).
(5) *Cool*.—Remove all sample bottles from the hot water bath (use appropriate gloves) and cool to approximately 60°C.
(6) *Protease treatment*.—Add 0.1 mL protease solution, *[2.2.4.1.3(5)]* with a positive displacement dispenser. Incubate at 60°C for 30 min.
(7) *pH adjustment*.—Add 4.0 mL of 2 M acetic acid *[2.2.4.1.3 (12)]* to each bottle and mix. This gives a final pH of approximately 4.3. If available carbohydrates are to be measured, remove an aliquot (0.5 mL) at this point and proceed to step *[2.2.4.4.4(1)]*.

2.2.4.1.7 Determination of HMWDF (IDF + SDFP)

(1) *Precipitation of high-molecular-weight soluble dietary fibre (HMWSDF; SDFP)*.—To each sample, add 1 mL of 100 mg/mL internal standard solution, *[2.2.4.1.3(8)]* then add 192 mL (measured at room temperature) of 95% v/v EtOH (or IMS), *[2.2.4.1.3(1)]* preheated to 60°C, and mix thoroughly. Allow the precipitate to form at room temperature for 60 min.
(2) *Filtration setup*.—Tare crucible containing Celite to nearest 0.1 mg. Wet and redistribute the bed of Celite in the crucible, using 15 mL of 78% v/v EtOH (or IMS), *[2.2.4.1.3(2)]* from wash bottle. Apply suction to crucible to draw Celite onto fritted glass as an even mat.
(3) *Filtration*.—Using vacuum, filter precipitated enzyme digest, *[2.2.4.1.7(1)]* through crucible. Using a wash bottle with 78% v/v EtOH or IMS *[2.2.4.1.3(2)]* quantitatively transfer all remaining particles to crucible. Retain and pool filtrate and washings, *[2.2.4.1.7(3)]* and *[2.2.4.1.7(4)]* respectively, for determination of low-molecular-weight soluble dietary fibre (SDFS), *[2.2.4.2.2(1)]*.
(4) *Wash*.—Using a vacuum, wash residue successively with two 15 mL portions of the following: 78% v/v EtOH or IMS; 95% v/v EtOH or IMS; acetone.

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(5) Dry crucibles containing residue overnight in 105°C oven.
(6) *Cool crucible*.—Cool in desiccator for approximately 1 h. Weigh crucible containing dietary fibre residue and Celite to nearest 0.1 mg. To obtain residue mass, subtract tare weight, *i.e.* weight of dried crucible and Celite.
(7) *Protein and ash determination*.—Analyse the residue from one crucible for protein, and the second residue of the duplicate for ash. Perform protein analysis on residue using Kjeldahl or combustion methods. Exercise caution when using a combustion analyser for protein in the residue. Celite volatilised from the sample can clog the transfer lines of the unit. Use 6.25 factor for all cases to calculate milligrams of protein. For ash analysis, incinerate the second residue for 5 h at 525°C. Cool in desiccator and weigh to nearest 0.1 mg. Subtract crucible and Celite weight to determine ash.

2.2.4.1.8 Calculations for HMWDF (IDF + SDFP)

(1) Blank (B, mg) determination.

$$B = [(\mathrm{BR1} + \mathrm{BR2})]/2 - P_B - A_B \quad (1)$$

where BR1 and BR2 = residue mass (mg) for duplicate blank determinations, respectively, and P_B and A_B = mass (mg) of protein and ash, respectively, determined on first and second blank residues.

(2) HMWDF (IDF + SDFP).

$$\mathrm{HMWDF(mg/100g)} = \{[(\mathrm{R1} + \mathrm{R2})/2 - P - A - B]/(\mathrm{M1} + \mathrm{M2})/2\} \times 100 \quad (2)$$

where R1 = residue mass 1 from M1 in mg; R2 = residue mass 2 from M2 in mg; M1 = test portion mass 1 in g; M2 = test portion mass 2 in g; A = ash mass from R1; P = protein mass from R2.

Data were analysed using an Excel calculator (Supporting Information Fig. S3).

2.2.4.2 Determination of SDFS

2.2.4.2.1 Introduction

Proper deionisation is an essential part of obtaining quality chromatographic data on SDFS. To obtain familiarity regarding the appearance of salt peaks in the SDFS chromatograms, dissolve 10 mg sodium chloride into 9 mL of deionised water and add 1 mL of glycerol internal standard, *[2.2.4.1.3(8)]* and proceed to *[2.2.4.2.2(4)]* at 'Transfer the solution from *[2.2.4.2.2(3)]* to a 10 mL disposable syringe…'. To ensure the resins being used are of adequate deionizing capacity, dissolve 50 mg sodium chloride in 9 mL of deionised water. Add 1 mL 100 mg/mL of LC internal standard, *[2.2.4.1.3(8)]* and proceed to step *[2.2.4.2.2(3)]* at 'Transfer 2 mL of this solution…'. The liquid chromatogram of this solution should show no peaks in the time range corresponding to carbohydrates of DP $\geq$ 3.

2.2.4.2.2 Procedure

(1) *Filtrate recovery, deionisation, and LC analysis*.—Set aside the filtrate from one of the sample duplicates, *[2.2.4.1.7 (3)/ 2.2.4.1.7(4)]* to use in case of spills or if duplicate SDFS data are desired. Transfer one-half of filtrate *[2.2.4.1.7(3)/ 2.2.4.1.7(4)]* of the other sample duplicate to a 500 mL evaporator flask and concentrate with a rotary evaporator to dryness at 50°C.
(2) *Dissolution of SDFS*.—Add 5 mL of distilled water to the evaporator flask and swirl the flask for approximately 2 min to dissolve the sample. Transfer the solution to a sealable 20 mL polypropylene tube.
(3) *Deionisation of sample*.—Transfer 2 mL of this solution to the top of a Bio-Rad Econo-Pac™ disposable chromatographic column (Cat. No. 732-1010) containing approximately 4 g Ambersep 200 (H^+)™ and approximately 4 g Amberlite FPA53 (OH^-)™ resins overlaid with a porous frit *[2.2.4.1.3(18)]*. Connect the column to a Gilson Minipuls® Evolution pump (http://www.gilson.com/en/Pipette/Products/53.229) or similar, and set the pump to give a column elution rate of 1 mL per minute. Collect the eluate in a 250 mL round-bottom, rotary evaporator flask. When the liquid level has dropped to the level of the porous frit, rinse the sample in with 2 mL distilled water and allow this to percolate into the resin collecting the eluate. Then add approximately 20 mL of distilled water to the top of the column and allow this to elute through the column into the 250 mL round-bottom flask at approximately 1 mL/min. Collect the eluate until the liquid level reaches the top of the resin in the column. Evaporate the solution to dryness at 50°C. Add 2 mL of distilled water and dissolve the sample by swirling the flask by hand for approximately 2 min. Transfer the sample into a polypropylene storage tube using a Pasteur pipet.

Alternate Procedure 1: Apply 2.5 mL of sample to the mixed bed resin and allow this to percolate into the resin and elute at approximately 1 mL/min using a mini-clamp to control flow rate. Add 2 mL of water and allow this also to percolate through the column at the same rate. Discard the eluate and washing (~4.5 mL). Add 20 mL of water to the top of the column and allow this to elute at approximately 1 mL/min. Collect this eluate and apply directly to the HPLC column after filtering as described in *[2.2.4.2.2(4)]*. [The sample has been diluted approximately eightfold].

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Alternate procedure 2: If online simultaneous deionisation and quantitation are carried out as per AOAC Method 2011.25, cation and anion exchange guard columns, H^+ and CO_2^{3-} forms, respectively (Bio-Rad Laboratories, Cat. No. 125-0118) with guard column holder (Bio-Rad Laboratories, Cat. No. 125-0139) to hold the two guard column cartridges in series (cation cartridge preceding anion cartridge) are used.

(4) *Preparation of samples for LC analyses.*—Transfer the solution from *[2.2.4.2.2(3)]* to a 10 mL disposable syringe, *[2.2.4.1.2(29)]* and filter through a 0.45 μm filter, *[2.2.4.1.2(27)]* Use a 100 μL LC glass syringe, *[2.2.4.1.2(30)]* to fill the 50 μL injection loop on the LC, *[2.2.4.1.2(21)]*. Perform this analysis in duplicate.

(5) *Determine the response factor forD-glucose.*—Because D-glucose provides an LC RI response equivalent to the response factor for the NDO that make up SDFS, the LC is calibrated using D-glucose, and the response factor is used for determining the mass of SDFS. Use a 100 μL LC syringe, *[2.2.4.1.2(30)]* to fill the 50 μL injection loop for each standard internal standard/D-glucose solution, *[2.2.4.1.3 (9)]*. Inject in duplicate onto TSK-GELR G2500PWXL® gel permeation columns or Waters Sugar-Pak® column.

(a) *Internal standard method.*—Obtain the values for the peak areas of D-glucose and internal standard from the three chromatograms. The reciprocal of the slope obtained by comparing the ratio of peak area of D-glucose/peak area of internal standard (*y*-axis) to the ratio of the mass of D-glucose/mass of internal standard (*x*-axis) is the response factor. Determine the average response factor (~0.8 for glycerol).

$$\text{Response factor (Rf)} = (\text{PA-IS})/(\text{PA-Glu}) \times (\text{Wt-Glu}/\text{Wt-IS}) \quad (3)$$

where PA-Glu = peak area D-glucose; PA-IS = peak area internal standard; Wt-Glu = mass of D-glucose (mg) contained in 1 mL of standard (5, 10, or 20 mg); Wt-IS = mass of internal standard (mg) contained in 1 mL of standard (10 mg glycerol)

(b.) *External standard method.*—Obtain the values for the peak areas of D-glucose from the three chromatograms. Determine the average response factor:

$$\text{Response factor (Rf)} = (\text{Wt-Glu})/(\text{PA-Glu}) \quad (4)$$

where PA-Glu = peak area D-glucose; Wt-Glu = mass of D-glucose (mg) contained in 1 mL of standard (5, 10 or 20 mg).

(6) *Calibrate the area of chromatogram to be measured for SDFS.*—Use a 100 μL LC syringe, *[2.2.4.1.2(30)]* to fill the 50 μL injection loop with retention time standard, *[2.2.4.1.3(7)]*. Inject in duplicate. Determine demarcation point between DP 2 and DP 3 oligosaccharides (disaccharides sucrose and maltose vs. higher oligosaccharides).

(7) *Determine peak area of SDFS (PA-SDFS) and internal standard (PA-IS) in chromatograms of sample extracts.*—Inject sample extracts, *[2.2.4.2.2(3)]* on LC system. Record area of all peaks of DP greater than the DP 2/DP 3 demarcation point as PA-SDFS. Record the peak area of internal standard as PA-IS.

2.2.4.2.3 Calculations for SDFS

(1) Internal standard method.

$$\text{SDFS(mg/100mg)} = \text{Rf} \times (\text{Wt-IS, mg}) \times (\text{PA-SDFS})/(\text{PA-IS}) \times 100/\text{M} \quad (5)$$

where Wt-IS is mg of internal standard contained in 10 mL internal standard solution pipetted into sample filtrate; PA-SDFS is the peak area of the low-molecular-weight soluble dietary fibre; PA-IS is the peak area of the internal standard; and M is the test portion mass M1 or M2 of the sample whose filtrate was concentrated and analysed by LC.

(2) External standard method.

$$\text{SDFS(mg/100mg)} = \text{Rf} \times (\text{PA-SDFS}) \times 100/\text{M} \quad (6)$$

where PA-SDFS is the peak area of the low-molecular-weight soluble dietary fibre; M is the test portion mass M1 or M2 of the sample whose filtrate was concentrated and analysed by LC.

2.2.4.3 Calculation of total dietary fibre

$$\text{TDF, \%} = (\text{HMWDF} + \text{SDFS}/1000) \quad (7)$$

Data were analysed using an Excel calculator (Supporting Information Fig. S3).

2.2.4.4 Measurement of available carbohydrates (ACH)

2.2.4.4.1 Principle

Non-resistant starch and maltodextrins are hydrolysed to D-glucose by the action of PAA and AMG in the TDF procedure. A sample (0.5 mL) of the reaction solution after pH adjustment with acetic acid is removed and centrifuged in a microfuge, and 0.2 mL is removed and diluted with sodium maleate buffer (pH 6.2). An aliquot of this solution is incubated with a mixture of sucrase/maltase

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plus β-galactosidase to hydrolyse sucrose and traces of maltose (if present in the sample) to D-fructose and D-glucose; and lactose to D-glucose and D-galactose. This mixture is analysed for D-glucose and D-fructose using hexokinase plus glucose 6-phosphate dehydrogenase, followed by phosphoglucose isomerase.

2.2.4.4.2 Apparatus

(1) Spectrophotometer set at 340 nm.
(2) Disposable plastic cuvettes (1 cm light path, 3.0 mL).
(3) Micro-pipettors, *e.g.* Gilson Pipetman® (20 and 100 μL).

- Positive displacement pipettor, *e.g.* Eppendorf Multipette® with 5.0 mL Combitip® (to dispense 0.2 mL aliquots)
- with 25 mL Combitip® (to dispense 2.0 mL aliquots).

(4) Vortex mixer (*e.g.* IKA® Yellowline Test Tube Shaker TTS2).

2.2.4.4.3 Reagents

(1) *Sodium maleate buffer (100 mM, pH 6.2) containing BSA (0.5 mg/mL) and sodium azide (0.02% w/v) as a preservative.* Dissolve 11.6 g of maleic acid in 800 mL of distilled water and adjust the pH to 6.2 with 4 M (160 g/L) NaOH solution. Add 0.5 g BSA and 0.2 g of sodium azide and dissolve. Adjust the volume to 1 L. Stable for >12 months at 4°C.
(2) *Imidazole buffer (2 M, pH 7.6) plus magnesium chloride (100 mM) and sodium azide (0.02% w/v) as a preservative.* Use as supplied (see below). Stable for >2 years at 4°C.
(3) *$NADP^+$ (21 mg/mL) plus ATP (42 mg/mL).* Stable for > 2 years at −20°C.
(4) *Sucrase (200 U) plus β-Galactosidase suspension (8000 U)*; freeze-dried powder. Dissolve in 10.5 mL of water and store frozen between use. Stable for >2 years at 4°C.
(5) *Hexokinase (425 U/mL) plus D-glucose-6-phosphate dehydrogenase (212 U/mL) suspension in 3.2 M ammonium sulphate.* Use as supplied. Stable for >2 years at 4°C.
(6) *Phosphoglucose isomerase suspension (1,000 U/mL) in 3.2 M ammonium sulphate.* Use as supplied. Stable for >2 years at 4°C.
(7) *D-glucose plus D-fructose standard solution (0.2 mg/mL of each sugar).* Use as supplied. Stable for >2 years at 4°C.

[Reagents (2)–(7) are available from Megazyme International Ireland].

2.2.4.4.4 Procedure

(1) Accurately transfer 0.5 mL of the solution recovered at *[2.2.4.1.6(7)]* of the TDF procedure to a microfuge centrifuge tube and centrifuge at 12 000 rpm for 2 min. Transfer 0.2 mL of the supernatant to 10 mL of 100 mM sodium maleate buffer (pH 6.2) *[2.2.4.4.3(1)]* (overall dilution of 51-fold) and mix well.
(2) Perform all other incubations as described in Fig. 4. Lactose is hydrolysed to D-glucose plus D-galactose by β-galactosidase and sucrose is hydrolysed to D-glucose and D-fructose by the sucrase/maltase enzyme. This enzyme has no action on low DP fructo-oligosaccharides such as kestose, kestotetraose, kestopentaose or oligosaccharides released on controlled hydrolysis of inulin by *endo*-inulinase (as in Raftilose P-95®).

2.2.4.4.5 Calculations

Determine the absorbance difference ($A_2 - A_1$) for both blank and sample. Subtract the absorbance difference of the blank from the absorbance difference of the sample, thereby obtaining ΔAD-glucose.

Wavelength: 340 nm
Cuvette: 1 cm light path (glass or plastic)
Temperature: ~ 25°C
Final volume: 2.52 mL (D-glucose)
2.54 mL (D-fructose)
Sample solution: 4-80 μg of D-glucose plus D-fructose per cuvette (in 0.10-2.00 mL sample volume)
Read against air (without a cuvette in the light path) or against water

Pipette into cuvettes	Blank	Sample
sample in maleate buffer (pH ~ 6.2)	-	0.20 mL
solution 2.2.4.4.3(4) (sucrase + β-galac)	-	0.10 mL
Ensure that all of the solutions are delivered to the bottom of the cuvette. Mix the contents by gentle swirling, cap the cuvettes and incubate them at ~ 25°C for 60 min. Add:		
distilled water (at ~ 25°C)	2.30 mL	2.00 mL
solution 2.2.4.4.3(2) (imidazole buffer)	0.10 mL	0.10 mL
solution 2.2.4.4.3(3) ($NADP^+$/ATP)	0.10 mL	0.10 mL
Mix*, read the absorbances of the solutions (A_1) after approx. 3 min and start the reactions by addition of:		
suspension 2.2.4.4.3(5) (HK/G6P-DH)	0.02 mL	0.02 mL
Mix*, read the absorbances of the solutions (A_2) at the end of the reaction (approx. 5 min). If the reaction has not stopped after 5 min, continue to read the absorbances at 2 min intervals until the absorbances remain the same over 2 min**. Then add:		
suspension 2.2.4.4.3(6) (PGI)	0.02 mL	0.02 mL
Mix*, read the absorbances of the solutions (A_3) at the end of the reaction (approx. 8-10 min).		

* for example with a plastic spatula or by gentle inversion after sealing the cuvette with a cuvette cap or Parafilm®.

** if the absorbance continues to increase, this may be due to effects of colour compounds or enzymes in the sample. These interfering substances may be removed during sample preparation.

Figure 4. Procedure for the measurement of D-glucose and D-fructose as released from non-resistant starch, maltodextrins, sucrose and lactose plus free D-glucose and D-fructose.

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Determine the absorbance difference ($A_3 - A_2$) for both blank and sample. Subtract the absorbance difference of the blank from the absorbance difference of the sample, thereby obtaining $\Delta A_{\text{D-fructose}}$.

The concentration of D-glucose and D-fructose (g/100 g) can be calculated as follows:

$$c = \frac{V \times \text{MW}}{\epsilon \times d \times v} \times F \times \frac{\text{EV}}{1000} \times \Delta A \times 100 \quad (8)$$

where V = final volume [mL], MW = molecular weight of D-glucose or D-fructose [g/mol], ϵ = extinction coefficient of NADPH at 340 nm = 6300 [$\text{l} \times \text{mol}^{-1} \times \text{cm}^{-1}$], d = light path [cm], v = sample volume [mL], F = dilution factor (usually 51-fold), EV = final sample extract volume (= 48.1 ± 0.05 mL). Empirically, it was found that 1 g of starch, D-glucose or sucrose in aqueous solution occupied a volume of 0.55 ± 0.05 mL and 100 = conversion of results to g/100 g.

It follows for D-glucose (g/100 g):

$$c = \frac{2.52 \times 180.16}{6300 \times 1 \times 0.2} \times \times 51 \times \frac{48.7}{1000} \times \Delta A_{\text{D-glucose}} \times 100 = 89.49 \times \Delta A_{\text{D-glucose}} \quad (9)$$

for D-fructose (g/100 g):

$$c = \frac{2.52 \times 180.16}{6300 \times 1 \times 0.2} \times 51 \times \frac{48.7}{1000} \times \Delta A_{\text{D-fructose}} \times 100 = 90.20 \times \Delta A_{\text{D-fructose}} \quad (10)$$

ACH (g/100 g):

$$c = \text{D-glucose (g/100 g)} + \text{D-fructose (g/100 g)}. \quad (11)$$

Data were analysed using an Excel calculator (Supporting Information Fig. S4).

3 Results and Discussion

In the application of AOAC Methods 2009.01 and 2011.25 over the past few years to the analysis of a range of food products and fibre ingredients, several problems/challenges were identified. These problems include the extended incubation time with the PAA plus AMG used in the assay procedure (16 h), compared to the likely residence time of food in the small intestine (4 ± 1 h) [24-33]; underestimation of FOS due to lack of separation of F3 from disaccharides on chromatography on the Waters Sugar-Pak® column; underestimation of phosphate crosslinked starch (RS_4); production and persistence of branched maltodextrins on hydrolysis of non-resistant starches; and, the need for the preservative sodium azide in the buffer employed.

In research performed by Englyst et al. [10] on readily digested starch (RDS), slowly digested starch (SDS) and resistant starch (RS) an incubation time of just 2 h was used to distinguish RDS + SDS from RS. In these experiments, the concentration of PAA was 19 KU BPU [= 7 K Ceralpha Units (KCU)] and of AMG was 1.5 KU per incubation (with 1 g of sample). All other reported analytical procedures for RS [22, 23, 40–46] employ much longer incubation times (~16 h) but with much lower concentrations of PAA and AMG. For example, McCleary and Monaghan [22] employed 0.12 KCU of PAA/0.1 g sample (*i.e.* 1.2 KCU/1 g sample), and 0.012 KU of AMG/0.1 g sample (*i.e.* 0.12 KU/1 g sample) in the incubations. In the analysis of a limited set of samples, similar RS values were obtained for the methods employing the 16 h incubation [22, 23, 40–46] to those obtained with the Englyst et al. [10] method, and were similar to those obtained through ileostomy studies [46]. To explain these differences, a set of experiments were designed to study the effects of enzyme concentration, time of incubation with PAA plus AMG, and incubation conditions on hydrolysis of a range of starch samples having quite different properties.

3.1 Hydrolysis of non-resistant starch—optimisation of incubation conditions

The relative rates of hydrolysis of various starch samples under conditions similar to those described by Englyst et al. [10] in centrifuge tubes with glass balls and guar gum was studied. In these experiments, either sodium acetate buffer (100 mM, pH 5.2) or sodium maleate buffer (100 mM, pH 6.0) with varying levels of PAA (1.5–24 KU Ceralpha/incubation) and AMG (0.34–3.4 KU/incubation) were employed. As shown in Fig. 5, in sodium maleate buffer (100 mM, pH 6.0) the maximal rate of hydrolysis (release of free glucose) of three quite different starch samples namely, Hylon VII® (a high amylose maize starch) (a and d), regular maize starch (RMS) (b and e), and native potato starch (NPS) (c and f) was achieved at a PAA concentration of 6 KU/incubation and AMG of 1.7 KU per incubation. Rates did not increase if the levels of either PAA or AMG used were above these concentrations. The same results were obtained when incubations were performed at pH 5.2 (a pH which is less physiologically significant).

The rates of hydrolysis of a) Hylon VII®, b) RMS, c) NPS and d) Fibersym® (a phosphate crosslinked starch; RS_4) under conditions very similar to those described by Englyst et al. [10] (pH 5.2, employing; (I). glass balls and guar in the incubation tube) (Englyst+M+G); (II) the same conditions but without the glass balls or guar from the incubation tubes (Englyst−M−G) and, (III) under the conditions as described in AOAC method 2002.02 (Resistant Starch; pH 6.0), are shown in Fig. 6. The rates of hydrolysis of the diverse starches in the presence or absence of the glass balls plus guar are very similar, indicating that potential grinding by

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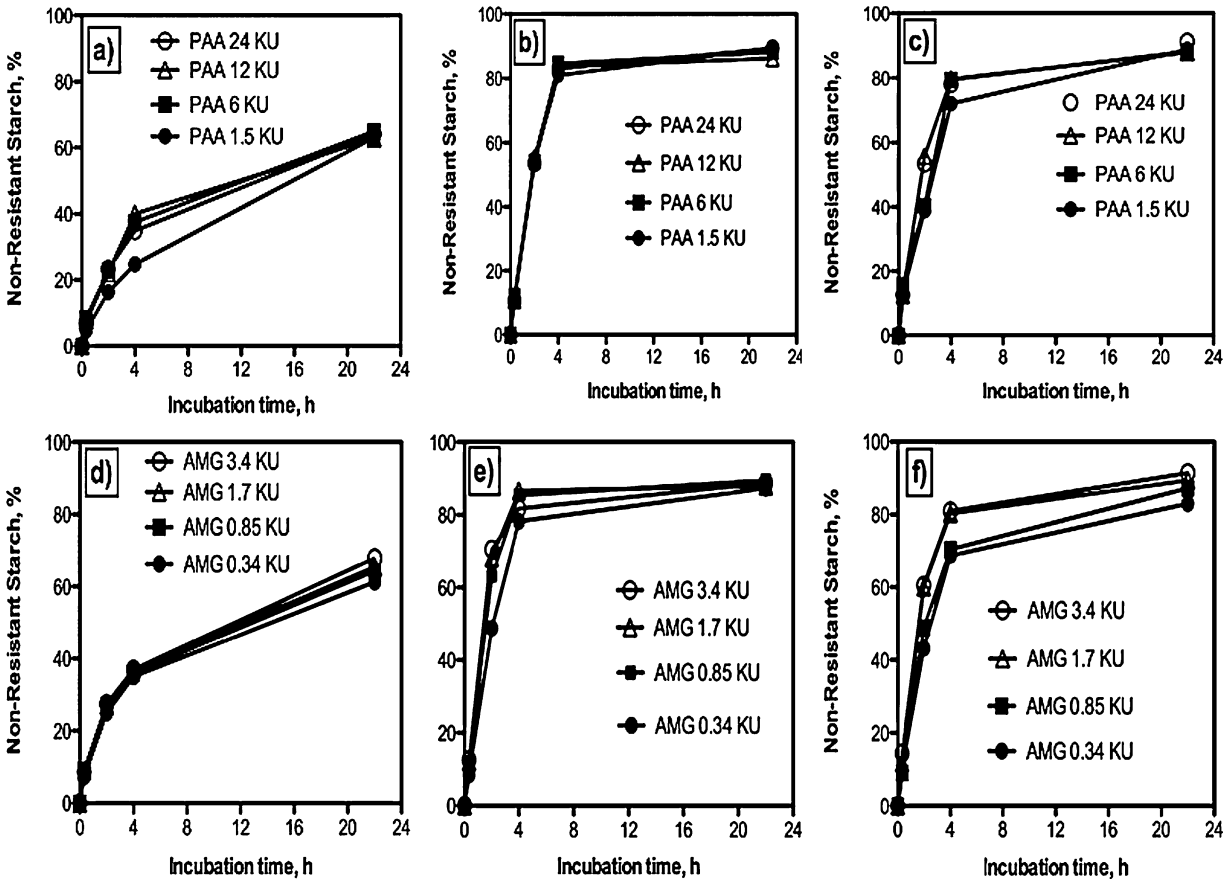

Figure 5. The effect of the concentration of PAA and of AMG on the extent of hydrolysis (release of free D-glucose, recorded as non-resistant starch) of Hylon VII, regular maize starch (RMS) and native potato starch (NPS) under conditions similar to those described by Englyst et al. [10] in a shaking water bath with glass marbles and guar gum in the incubation tube, at 37 °C, but at pH 6.0 (instead of pH 5.2). Samples 'a–c' were incubated with 1.7 KU AMG and varying levels of PAA. Samples 'd–f' were incubated with 12 KU of PAA and varying levels of AMG. Samples 'a' and 'd' are Hylon VII; samples 'b' and 'e' are RMS; samples 'c' and 'f' are NPS. Samples were analysed for free D-glucose as described in Materials and methods 2.2.2.1.

the glass balls appears to make no significant difference to the rates of hydrolysis of the starches.The extents of hydrolysis of RMS and NPS after 16 h under the conditions described in AOAC Method 2002.02, are similar to those obtained using conditions as described by Englyst et al. [10] (with or without marbles and guar gum) after just 4 h. However, for Hylon VII® (high amylose maize starch) and particularly for Fibersym® (phosphate crosslinked starch), the extent of hydrolysis by the Englyst et al. [10] method after 4 h was significantly less than that obtained under the conditions of AOAC Method 2002.02 after 16 h. An incubation time of 4 h is consistent with the time of residence of food in the small intestine as reported in numerous independent studies [24–33]. Of course, in vivo conditions in the small intestine cannot be perfectly replicated under in vitro incubation conditions, but in the method described herin, the pH, temperature and incubation time are consistent with likely in vivo conditions. Also, the level of PAA is saturating, and while it is not feasible to routinely employ the mucosal α-glucosidase/glucoamylase enzyme complex from the brush boarder lining of the small intestine, these are simulated as closely as possible with saturating levels of fungal AMG (glucoamylase) under incubation conditions where the sample is continually suspended by stirring or shaking.

The determination of dietary fibre is most conveniently performed under conditions where all operations, including addition of ethanol/IMS, are performed in the same container to minimise sample losses. Containers which satisfy these requirements are Fisherbrand® 250 mL soda glass, wide-mouthed bottles with polyvinyl-lined caps. These containers hold the required volume of liquids, and insoluble fibre samples are simple to remove with a rubber 'policeman' spatula. The rates of hydrolysis of a range of starch samples [a) Hylon VII®, b) wheat starch (WS), c) NPS and d) Fibersym®] by a mixture of PAA (6 KU/incubation) and AMG (1.7 KU/incubation) in maleate bufer (pH 6.0) [*2.2.4.1.3(10)*]

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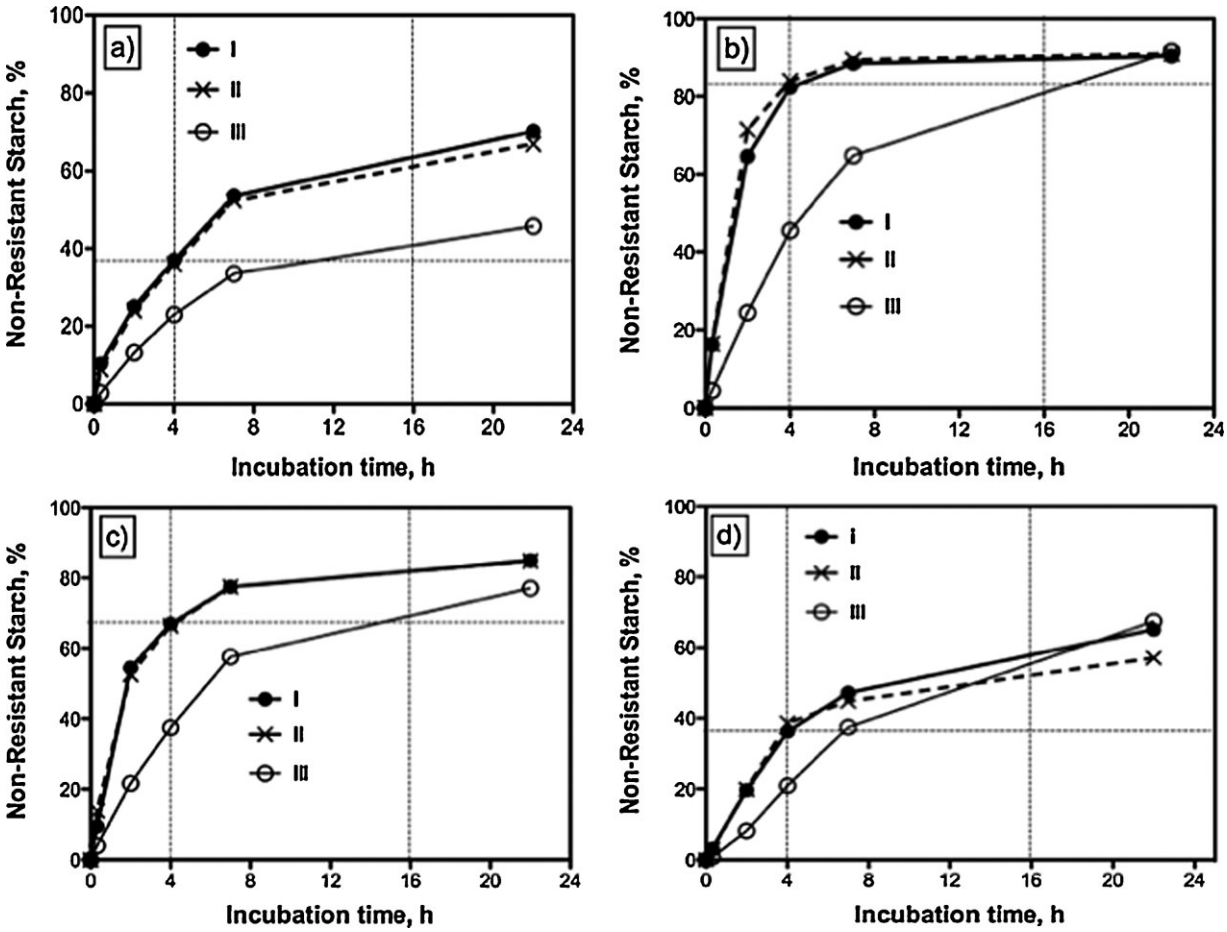

Figure 6. Time course hydrolysis of a) Hylon VII®, b) regular maize starch (RMS), c) native potato starch (NPS) and d) Fibersym® by a mixture of PAA (6 KU/assay) and AMG (1.7 KU/assay), otherwise under the conditions described by Englyst et al. [10] in a shaking water bath at pH 5.2 and 37°C; I) with glass marbles and guar gum in the incubation tube; II) with no marbles or guar gum in the incubation tubes. Alternatively, III) with the much lower levels of PAA and AMG according to AOAC Method 2002.02. Samples were analysed for free D-glucose as described in Materials and methods 2.2.2.1.

in Fisherbrand® bottles with shaking or stirring of the contents was compared to the rates obtained using polypropylene centrifuge tubes (with glass balls and guar) (Fig. 7). Clearly, up to an incubation time of 4 h, the values obtained for non-resistant starch for all four incubation methods are very similar.

On the basis of these experiments described above, all further incubations were performed in Fisherbrand® bottles with stirring by a magnetic stirrer bar added into the bottle. The rates of hydrolysis of Hylon VII® (a and d) RMS (b and e) and Fibersym® (c and f) under these conditions, with varying levels of PAA and AMG at pH 6, is shown in Fig. 8. PAA is saturating above 3 KU per incubation, and AMG is saturating above 1.7 KU per incubation. Thus, the final incubation conditions that were chosen for analysis of samples for DF involve suspension of 1 g of sample in 40 mL of maleate buffer (pH 6) *[2.2.3.1.3(10)]* and incubation with stirring at 170 rpm and 37°C with PAA (6 KU) and AMG (1.7 KU) for 4 h.

3.2 Recovery of NDO under the new DF incubation conditions

Since the concentrations of PAA and AMG in this modified incubation format are significantly higher than those used in AOAC TDF Methods 2009.01 and 2011.25, it was essential to demonstrate that there was no additional hydrolysis of NDO such as FOS, GOS, XOS, galactosyl-sucrose oligosaccharides, Fibersol 2® or Polydextrose®. HPLC patterns obtained on chromatography of these oligosaccharides on TSK-GELR G2500PWXL® gel permeation columns before and after incubation according to the recommended conditions described here [Method 2.2.4] are shown in Fig. 9 and Supporting Information Fig. S5. Recovery values for the original oligosaccharide; after incubation according to AOAC Method 2009.01; after incubation according to AOAC Method 2009.01 with the additional AMG incubation [8]; and after incubation under the current conditions (RINTDF) are summarised in Table 1. For all oligosaccharides except

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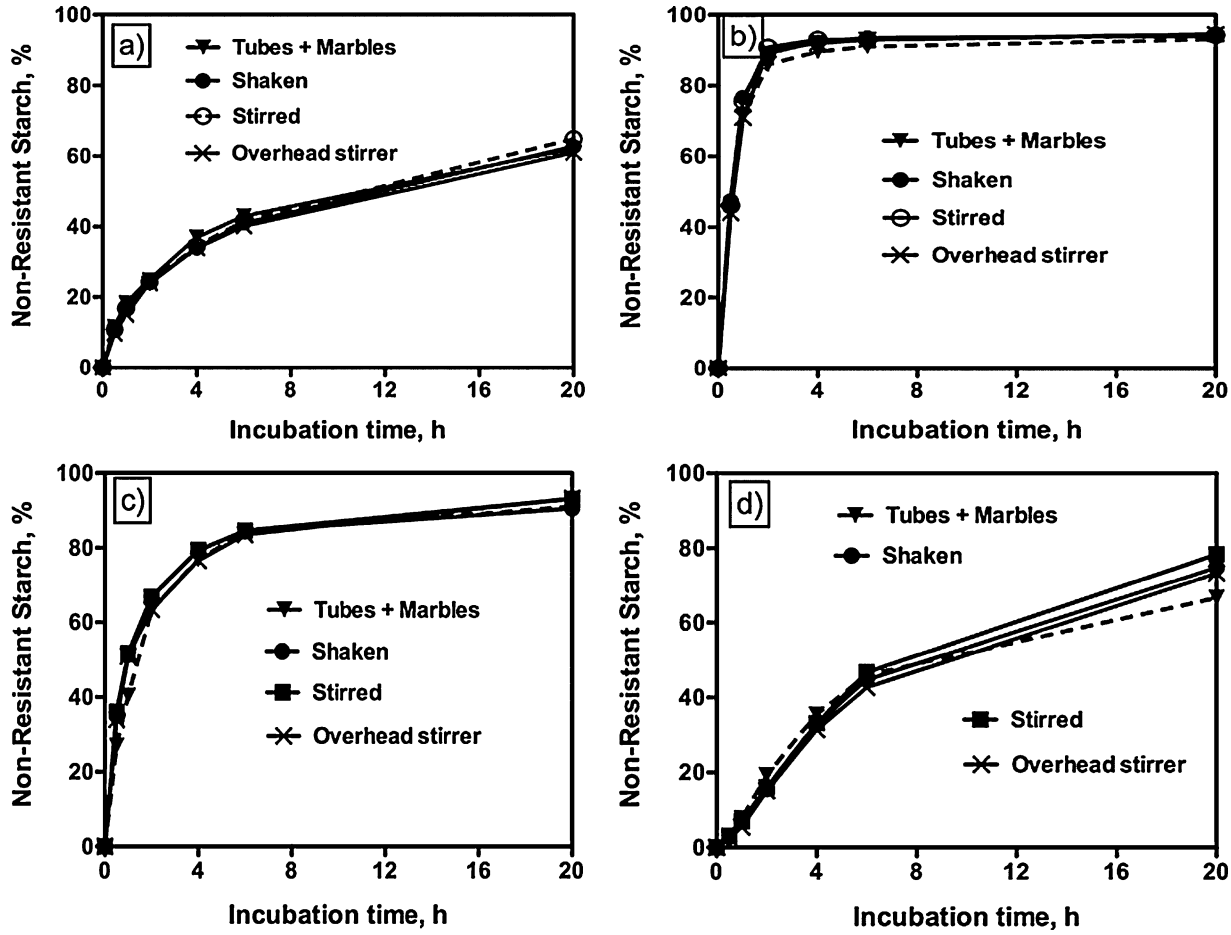

Figure 7. Time course hydrolysis of a) Hylon VII®, b) wheat starch (WS), c) native potato starch (NPS) and d) Fibersym® by a mixture of PAA (6 KU/assay) and AMG (1.7 KU/assay) under the following conditions; those described by Englyst et al. [10] in a shaking water bath at 37 °C, but at pH 6.0 with glass marbles and guar gum in the incubation tube (▼); in Fisherbrand® bottles with orbital shaking at 150 rpm (●); in Fisherbrand® bottles with stirring using a magnetic stirrer bar added to the bottle (■); and in Fisherbrand® bottles with stirring using a suspended magnetic stirrer bar (X). Samples were analysed for free D-glucose as described in Methods.

AdvantaFiber®, the amount of recovered oligosaccharide is very similar to that obtained using AOAC Method 2009.01. AdvantaFiber® is a mixture of isomalto-oligosaccharides and clearly these are readily hydrolysed under each of the incubation conditions described here, and hydrolysis increases as the level of PAA and AMG increases. This is consistent with the known hydrolysis of isomaltose and isomalto-oligosaccharides by the α-glucosidase/glucoamylase complex of the brush boarder lining of the small intestine [8, 38, 47].

3.3 Recovery of DF for the high molecular weight dietary fibre (HMWDF) controls

The results obtained using the current format for the control materials routinely used to check the purity of the enzymes employed in TDF assays are shown in Table 2. Values obtained for β-glucan, wheat starch, pectin, larch galactan and wheat arabinoxylan are in line with those obtained using AOAC Methods 2009.01 and 991.43, consistent with the high purity of the enzymes employed. A higher value was obtained for Hylon VII® (high amylose maize starch), more in line with values obtained for similar high amylose native starches in some studies [10, 43].

3.4 Desalting of NDO samples in preparation for HPLC

In the procedure described here, 2 mL of concentrated SDFS fraction is applied to a mixed bed resin containing 4 g Amberlite® FPA53 (OH^-) resin and 4 g Ambersep® 200 (H^+) resin and the sugars and internal standard are eluted with 2 + 20 mL of deionised water. This solution is then concentrated to dryness and re-dissolved in 2 mL of deionised water ready for HPLC. In an alternative procedure for preparing SDFS samples for HPLC {*[2.2.4.2.2(3)];*

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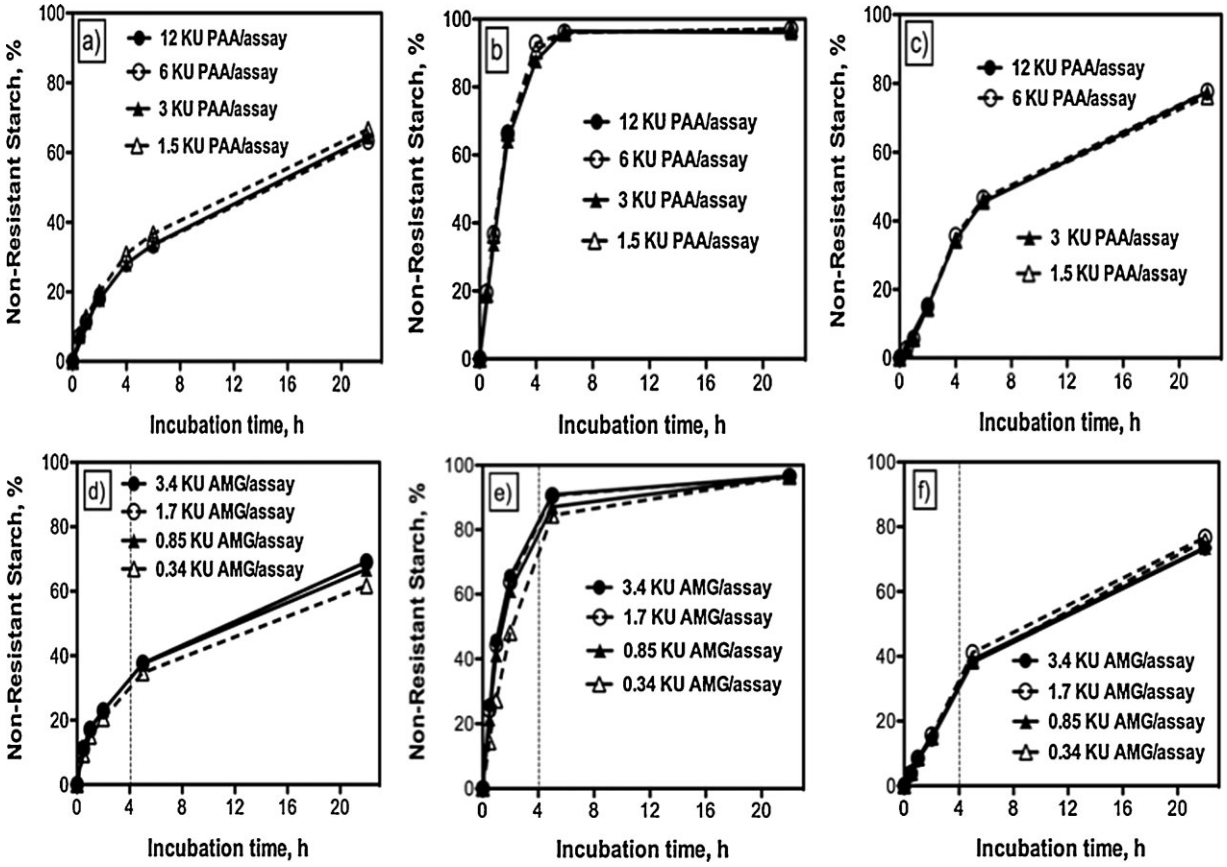

Figure 8. The effect of the concentration of PAA and of AMG on the extent of hydrolysis of Hylon VII® (a & d), RMS (b & e) and Fibersym® (c & f) in Fisherbrand® bottles with magnetic stirring at 170 rpm and at 37°C and pH 6.0. Samples were analysed for free D-glucose as described in Materials and Methods 2.2.2.A and 2.2.2.B. Sample a–c were incubated with 1.7 KU AMG and varying levels of PAA. Samples d–f were incubated with 6 KU PAA and varying levels of AMG.

'Alternate Procedure 1'}, 2.5 mL of sample was added to the mixed-bed resin at an elution rate of approximately 1 mL/min and washed into the bed with 2 mL of deionised water. This eluate (4.5 mL) was discarded. The resin was then eluted with 20 mL of deionised water and this solution was applied directly to the HPLC column without re-concentrating. As can be seen from Fig. 10, the determined ratio of FOS to glycerol is the same whether the sample is re-concentrated or not. The accuracy with non-concentrated samples may suffer for samples containing low levels of SDFS, but this remains to be determined.

3.5 Low molecular weight branched maltodextrins (LMWBMD) produced on hydrolysis of starch

On incubation of samples containing high levels of starch (*e.g.* rice and white bread) under the conditions described in AOAC Method 2009.01, small amounts of branched maltodextrins are produced as a by-product, that are very resistant to further hydrolysis by PAA or AMG. The heptasaccharide 6^3,6^5-di-α-D-glucosyl maltopentaose and the tetrasaccharide 6^3-α-D-glucosyl maltotriose are the major products as determined by GC-MS and NMR analysis of the purified oligosaccharides [8]. These compounds, particularly the heptasaccharide, are highly resistant to hydrolysis by PAA and AMG, but are readily hydrolysed by a mucosal α-glucosidase preparation from the small intestine of pig which is considered to be very similar to that from human small intestine [38]. Consequently, by definition they cannot be considered to be dietary fibre so they must be removed and thus not be included in the analytical value for SDFS. This can be achieved by inclusion of an additional incubation of the SDSF fraction with high levels of AMG at 60°C [8, 35, 36]. These conditions have been shown to give no additional hydrolysis of other NDO [8]. However, including yet another step in an already challenging assay procedure is not ideal. In the current procedure, the ratio of PAA and AMG was selected such that these oligosaccharides are simply not produced in the initial incubation step; meaning that the additional AMG incubation step is not required. This is clearly advantageous. HPLC on TSK-GELR G2500PWXL® gel permeation columns of the SDFS fractions obtained from pre-boiled rice using the current incubation format and that described in AOAC Method 2009.01 are shown in Fig. 11.

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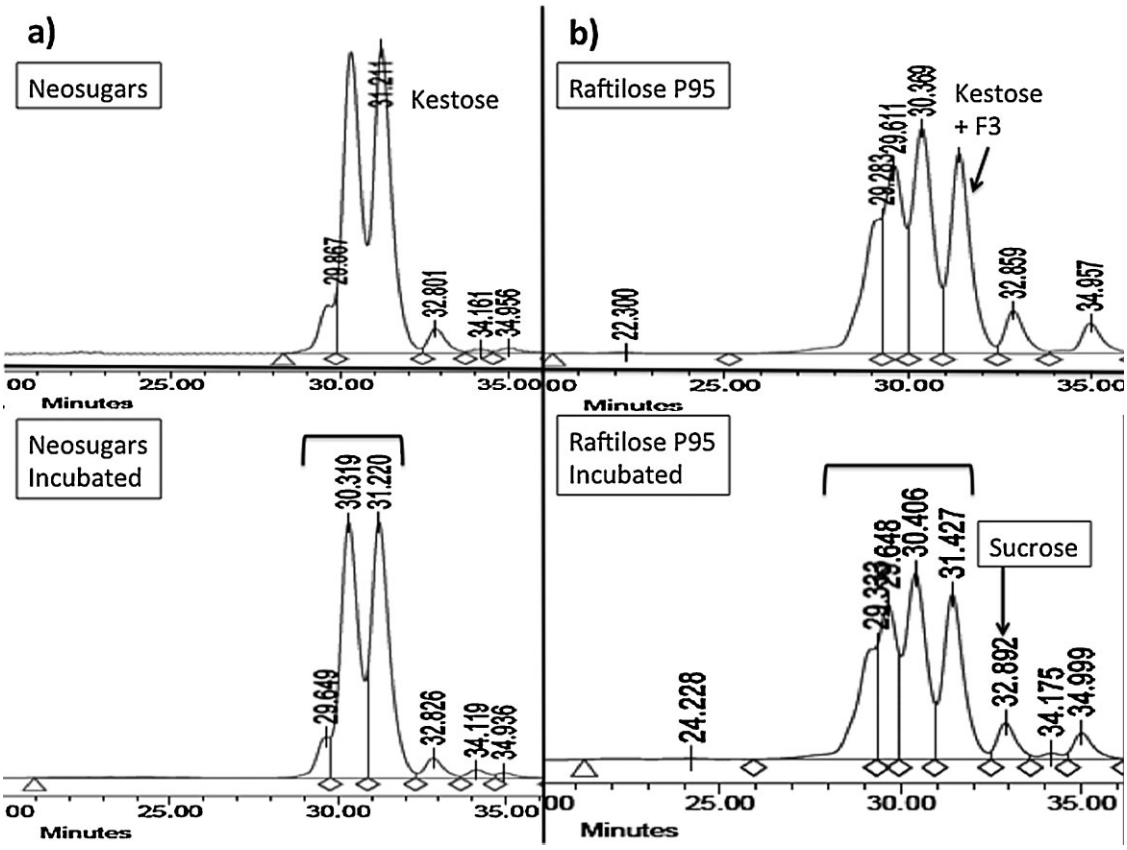

Figure 9. HPLC on TSK-GELR G2500PWXL® columns of Neosugars® and Raftilose® P-95 before and after incubation with PAA/AMG and protease according to the Integrated RINTDF method described in this paper [2.2.4].

Clearly, the resistant branched maltodextrins derived from starch are hydrolysed in the current incubation format, but not in the AOAC Method 2009.01 format. A similar observation was made with several other samples including cassava starch (Supporting Information Fig. S6), high amylose maize starch (see Table 3) and regular maize starch (not shown). Hydrolysis of 6^3,6^5-di-α-D-glucosyl maltopentaose and 6^3-α-D-glucosyl maltotriose under the

Table 1. Recovery of oligosaccharides of DP ≥ 3 in original samples and on incubation of the samples according to AOAC 2009.01, AOAC 2009.01 + extra AMG and the new format (RINTDF)[a)]

Sample	Recovery of Oligosaccharides of DP ≥ 3 as a percentage of total carbohydrate in the sample (% w/w)			
	Original Oligosaccharides	AOAC Method 2009.01[b)]	AOAC Method 2009.01 + AMG[b)]	New Format (RINTDF)
Neosugars®	93.0	92.9	92.7	92.8
Raftilose P95®	91.2	90.3	89.7	89.1
Polydextrose®	84.3	85.1	84.8	82.5
Fibersol 2®	88.5	83.4	84.9	82.4
Galacto-oligosaccharides	76.0	70.6	71.3	72.0
Xylo-oligosaccharides	78.0	78.6	77.7	76.2
Raffinose	99.0	99.0	99.0	98.0
AdvantaFiber®	65.4	29.0	16.3	10.8
Coupling Sugars®	43.8	–	–	0

a) Calculated from HPLC patterns as areas under the peaks for oligosaccharides of DP ≥ 3 as a percentage of combined area for all peaks from the sample.
b) From McCleary et al. [37]. Reprinted as by permission of AOAC International.

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Table 2. TDF values determined for a range of samples traditionally used as standards in TDF assays

Sample	AOAC Method 991.43[a)]	AOAC Method 2009.01[a)]	Current Method (RINTDF)
β-Glucan	98.0	96.0	96.2
Citrus Pectin	86.5	87.0	84.5
Larch galactan	83.5	84.0	86.0
Wheat arabinoxylan	95.0	94.5	96.0
Hylon VII® (high amylose maize starch)	29.3	46.5	58.8
Casein	0	0	0.2
Wheat starch	0.1	0.1	0.1

a) From McCleary [19]. (Reprinted as by permission of Springer publishers.)

conditions used in the current format was further studied by incubating 50 mg of each oligosaccharide under the standard incubation conditions. Both oligosaccharides were completely hydrolysed (Fig. 12), indicating that the concentrations of the enzymes, particularly AMG, are adequate to catalyse hydrolysis of the relatively resistant oligosaccharides to glucose.

3.6 HPLC chromatographic system, chromatography of FOS and internal standards

In the development of AOAC Method 2009.01 (an integrated procedure for the measurement of TDF), we chose to use Waters Sugar-Pak® HPLC columns with D-sorbitol as the internal standard. Glycerol was not used because the enzymes employed in the incubations contained glycerol as a stabiliser. It was subsequently found that certain commercially available FOS contain a trisaccharide, fructosyl-β-(2-1)-fructosyl-β-(2-1)-fructose (F3) at up to 15% w/w, which chromatographs on a Waters Sugar-Pak® HPLC column at the same point as the disaccharides, maltose, sucrose and lactose, which makes accurate measurement of FOS difficult. One solution to this problem involves hydrolysing the disaccharides in SDFS with a mixture of sucrase plus β-galactosidase (Megazyme cat. no. E-SUCRBG) and re-chromatographed the SDFS fraction to get a measure of

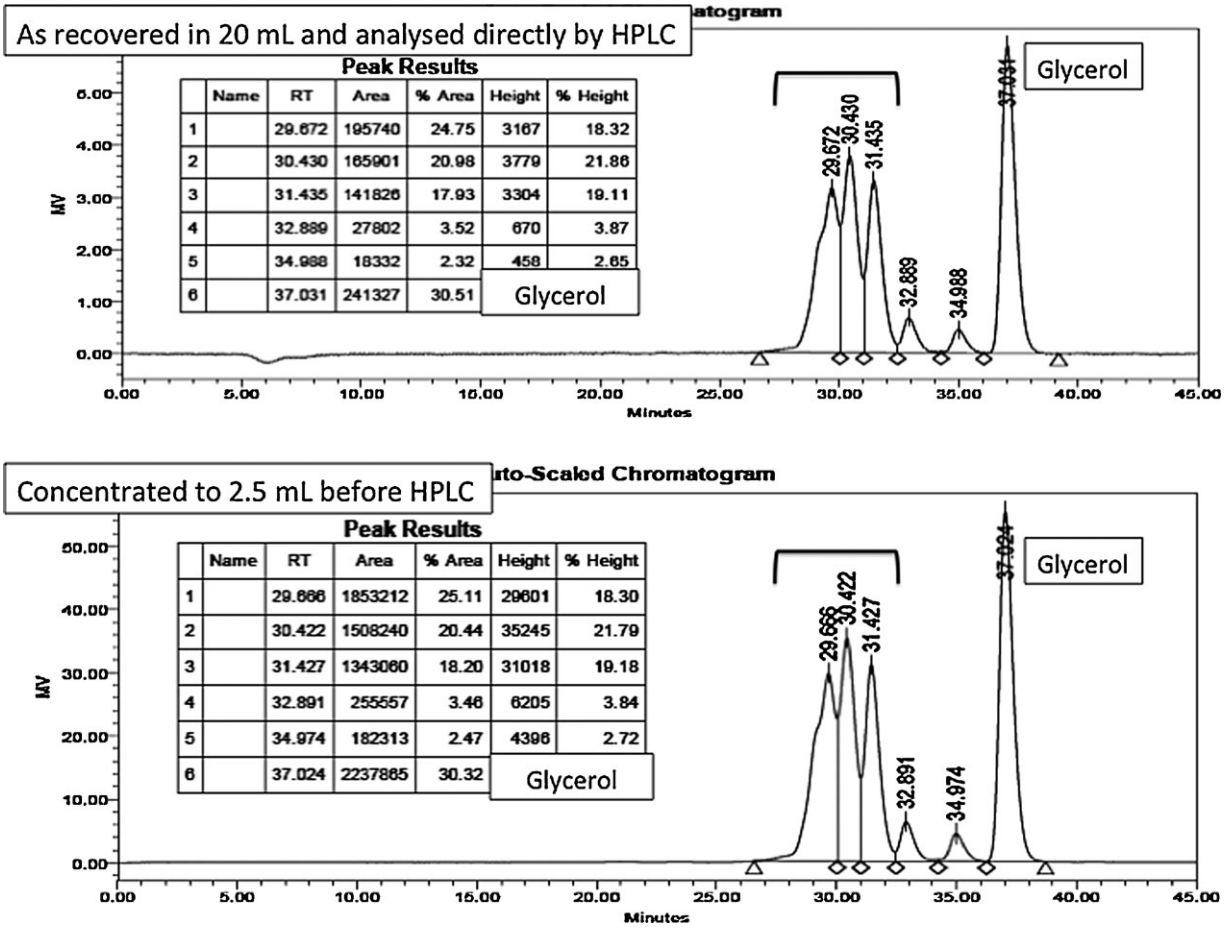

Figure 10. HPLC on TSK-GELR G2500PWXL® columns of the SDFS fraction obtained on incubation of Raftilose P95 (200 mg) plus glycerol (100 mg) recovered in 20 mL from ion-exchange column, either applied directly, or concentrated to dryness and redissolved in 2.5 mL of distilled water (Materials and methods *[2.2.4.2.2(3)]* Alternate Procedure 1).

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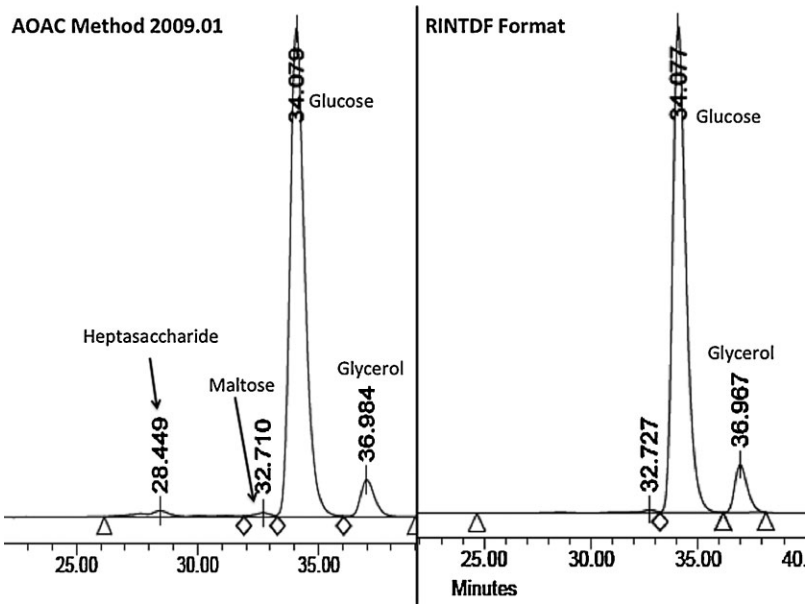

Figure 11. HPLC on TSK-GELR G2500PWXL® columns of the SDFS fraction obtained from Uncle Ben's Ready Rice incubated according to AOAC Method 2009.01 and the new DF assay format (RINTDF).

the F3. However, this additional incubation step can be avoided simply by employing TSK-GEL G2500PWXL® gel permeation columns, which give clear delineation between F3 and disaccharides (see Fig. 9). If TSK-GEL G2500PWXL® gel permeation chromatography is performed, D-sorbitol cannot be used as the internal standard because it elutes at the same point as glucose. Glycerol is the internal standard of choice, so enzymes employed in the assay must be free of glycerol. The enzymes recommended for this procedure are devoid of glycerol. PAA and AMG are available as a powder mixture (Megazyme cat. no. E-PAAMG) and are dissolved before use. Protease (Megazyme cat. no. E-BSPRPD) is also used in powder form. For general use, the Waters Sugar-Pak® HPLC column setup has the advantage of lower cost and quicker chromatography (approximately 20 min) compared to TSK-GEL G2500PWXL® gel permeation columns

Table 3. A comparison of dietary fibre values obtained for a range of starch containing samples using AOAC Method 2009.01 (plus additional AMG incubation step) with those obtained using the new assay format (RINTDF)

	AOAC Method 2009.01 + Extra AMG			New Format (RINTDF)		
Sample	HMWDF %	SDFS %	TDF (HMWDF + SDFS) %	HMWDF %	SDFS %	TDF (HMWDF + SDFS) %
Wholemeal bread	12.4	1.8	14.2	12.0	1.5	13.5
White bread	4.8	0.5	5.3	4.6	1.4	6.0
Oat bran	18.9	0.5	19.4	19.9	1.4	21.3
Weetabix	9.8	2.3	12.1	10.1	1.4	11.6
Kellogg All Bran	26.6	2.5	29.1	28.1	3.6	31.7
Whole wheat pasta	9.9	1.9	11.8	10.1	2.2	12.3
Chick peas (tinned)	18.0	1.3	19.3	17.6	2.1	19.7
Semi-ripe banana	30.2	0.9	31.1	30.2	1.4	31.6
Butter beans (tinned)	20.2	2.2	22.4	19.9	3.0	22.9
Sweet corn (tinned)	12.7	0.1	12.8	12.4	0.5	12.9
Garden peas (tinned)	29.1	1.3	30.4	29.1	2.2	31.3
Cabbage	28.9	1.1	30.0	26.5	1.3	27.8
Broccoli	28.1	0.4	28.5	29.7	0.6	30.3
Carrots	21.8	0.5	22.3	22.2	1.2	23.4
Fibersym®	28.6	1.1	29.7	59.2	1.0	60.2
Fiber Rite®	8.7	1.2	9.9	8.0	1.1	9.1
Hylon VII® (high amylose maize starch)	49.3	0.0	49.8	58.8	0.0	58.8
Polydextrose®	1.5	83.3	84.8	1.1	83.3	84.4

a) From McCleary et al. [8] (reprinted as by permission fo AACC).

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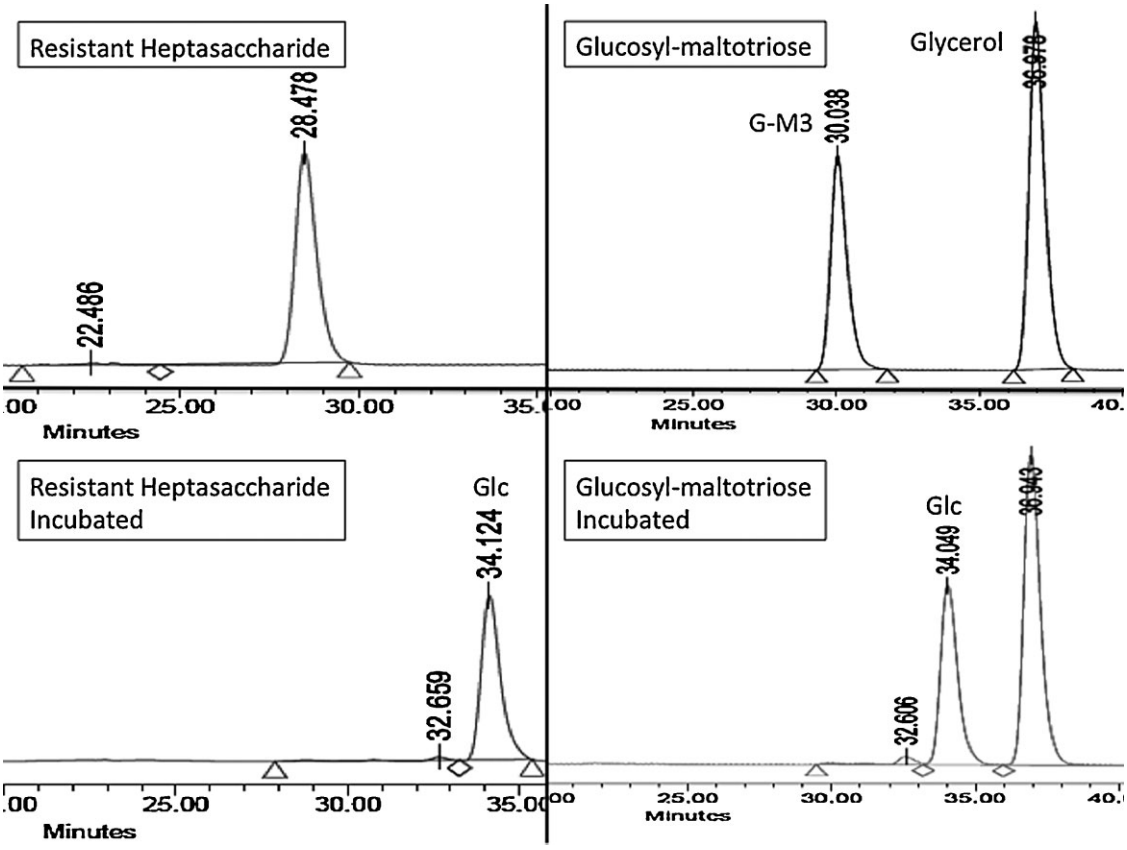

Figure 12. HPLC on TSK-GELR G2500PWXL® columns of the resistant heptasaccharide and 6^3-α-D-glucosyl maltotriose before and after incubation under the conditions used in the new TDF assay format (RINTDF).

(approximately 45 min). However, for unknown samples and particularly for samples known to contain FOS, the TSK-GEL G2500PWXL® gel permeation column setup is recommended.

3.7 Deletion of sodium azide from the sodium maleate buffer used in the DF procedure

Sodium azide is a very effective preservative and as such, is a poisonous chemical. Even though it is used at low concentrations (0.02% w/v) in the assay buffer, there are still concerns by some analysts in its routine use in analytical laboratories. In the original integrated DF procedures (AOAC Method 2009.01/2011.25), we included sodium azide in the buffer because we were concerned about microbial infection of the incubation mixture on incubation at 37°C for 16 h. In the RINTDF procedure with an incubation time of just 4 h, problems of microbial contamination are decreased considerably. To study this, glucose (1 g) was run through the incubation with PAA/AMG for up to 20 h and samples of the reaction solution were removed before addition of the PAA/AMG, and then at 1, 2, 4, 6 and 24 h and analysed for glucose. No decrease in the level of glucose, determined using GOPOD Reagent, was observed, indicating that there was no microbial consumption of D-glucose and by inference that microbial contamination was not a problem. In routine use, however, we recommend that either fresh buffer is used, or that before use, buffer is pre-sterilised by heated it to ~90°C in a boiling water bath or in a microwave oven.

3.8 Total dietary fibre content of a range of samples

Total dietary fibre values for a range of samples analysed with AOAC Method 2009.01 with the additional AMG incubation step [37] and with the new procedure (RINTDF) are shown in Table 3. Clearly, there is good agreement across a wide range of samples between the new format and AOAC 2009.01 (including the extra AMG incubation step). The two major exceptions noted were for Fibersym® (RS_4 phosphate crosslinked starch) and Hylon VII® (high amylose maize starch). The pattern of hydrolysis of these starches appears to be more dependent on incubation time rather than on enzyme concentrations alone. We consider that the values obtained with the new format more accurately reflect in vivo results because the incubation conditions more closely

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reflect those experienced in the human small intestine, *i.e.* 4 h residence time, 37°C, pH 6, PAA plus AMG (AMG simulating the mucosal glucoamylase/α-glucosidase complex). Also, the determined RS values obtained for a number of samples analysed under these conditions closely resemble values obtained in in vivo ileostomy studies.

Recently, Tanabe et al. [47] highlighted what they saw as problems with AOAC Method 2009.01 for the measurement of non-digestible oligosaccharides and suggested that the method could be improved by replacing AMG with porcine intestinal enzymes. The problem they identified was that sucrose, palatinose, panose, isomalto-oligosaccharides and 'Coupling Sugar®' were not hydrolysed (or not completely hydrolysed) in AOAC Method 2009.01, and thus led to inaccuracies. For clarity, disaccharides such as sucrose and palatinose are not included in dietary fibre determinations anyway, because DF as defined by Codex Alimentarius, comprises oligosaccharides of DP ≥ 3. In AOAC Method 2009.01, Coupling Sugar® is hydrolysed to sucrose and glucose, so is not measured as dietary fibre. Panose is incompletely hydrolysed in AOAC 2009.01, but is completely hydrolysed in the current format (RINTDF), and approximately 90% of an isomalto-oligosaccharide mixture (AdvantaFiber®; Table 1; Supporting Information 1) is hydrolysed to glucose and disaccharide. While PAA plus a porcine intestinal preparation may appear to be an ideal mixture to use in the assay, it is totally impractical as it is not possible to produce porcine intestinal preparation in sufficient quantities to allow its routine use in dietary fibre determinations. AMG is used as the best possible alternative to porcine intestinal preparation, and the results obtained with a mixture of this with PAA are very similar to that obtained with PAA and the porcine intestinal preparation (when considering oligosaccharides of DP ≥ 3).

3.9 Effect of cooking on the determined RS values for a range of starch samples

In RS and TDF analytical procedures, the samples analysed are quite often raw starches and milled cereal grains. But this is not how they are consumed. To obtain information on the effect of cooking on determined RS values, samples of starch containing materials (1 g) were suspended in maleate buffer and heated in a boiling water bath for 15 min (Materials and methods 2.2.3) and cooled to 37°C before proceeding with the incubation with PAA/AMG under standard conditions. Aliquots (4 mL) of incubation solution were added to 4 mL of ethanol and analysed for RS as described. From the results summarised in Table 4, it is clear that the analysed values for most samples are dramatically affected by cooking of the sample, as would be expected. The exceptions are Novelose 330® (retrograded high amylose maize starch) [48], Heinz baked beans, RTH rice and Uncle Ben's Ready white rice.

Table 4. Effect of pre-cooking[a)] of starch containing samples on the determined resistant starch values[b)] (Materials and methods 2.2.3)

Sample	Not pre-cooked	Pre-cooked (15 min at ~100°C)
Wheat starch (Sigma)	0.21	0.8
Regular maize starch (Lot 140206)	5.7	0.8
Hylon VII® (Lot 60901)	56.7	21.0
High amylose maize starch (Lot 60107)	46.6	20.7
Novelose 240® (National Starch)	44.3	29.4
Novelose 330® (National Starch)	39.3	41.0
Hi Maize 1043	48.1	30.7
ActiStar®	54.9	33.7
Amylose (potato)	33.6	33.4
Native potato starch (Avebe)	57.6	0.5
Heinz baked beans	4.5	3.7
Semi-ripe banana (Lot 61002)	26.9	0.3
RTH Rice (Mars corporation)	4.6	4.8
Uncle Ben's Ready white rice	3.8	3.8

a) Pre-cooked in a boiling water bath for 15 min.
b) All samples were analysed in duplicate.

The lack of change for these four samples following further cooking is most probably due to the fact that each of these samples was already pre-cooked as received. Clearly, the cooking process employed to produce the RTH and Ready white rice produce high levels of resistant starch. Changes in the levels of RS as a result of cooking must be considered when including specific resistant starch rich ingredients in various processed food products.

3.10 Available carbohydrates

Edible carbohydrates can conveniently be divided into high molecular weight dietary fibre (HMWDF), low-molecular-weight dietary fibre (LMWDF; NDO; SDFS) and available carbohydrates (ACH). In the assay procedure described here for measurement of dietary fibre, values for the ACH content of the sample can simply be obtained by analysing an aliquot of the hydrolysate as described in Materials and methods 2.2.4.4. In these studies, ACH is described as the sum of total D-glucose and D-fructose derived from non-resistant starch, maltodextrins, sucrose, free D-glucose and D-fructose and the D-glucose component of lactose. Essentially, complete hydrolysis of non-resistant starch and maltodextrins to D-glucose is achieved under the incubation conditions with PAA and AMG in the assay procedure. Sucrose, lactose and minor traces of maltose are hydrolysed to D-glucose, D-fructose and D-galactose as described in Materials and methods and summarised in Fig. 4. The total D-glucose, D-fructose and ACH values for a number of fruits, vegetables, cereals and food products are shown in Table 5. The ACH values were obtained simply by summing the

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values for the D-glucose and D-fructose. This assay measures the D-glucose component of lactose, but not the D-galactose. Because D-galactose has a glycemic index of approximately zero, it is usually not measured. However, if lactose and D-galactose values are required, they can be measured using a lactose assay kit (Megazyme cat. no. K-LACGAR).

Since most starches are readily hydrolysed, materials containing starch (*e.g.* cereals and some vegetables) have a high ACH content. In fact, pure starch has a D-glucose content of ~110%. This is because D-glucose in starch is in the anhydro form. Once released, a water molecule is added for each D-glucose, increasing the weight value by 180/162.

To obtain accurate measurement of ACH, the conditions of hydrolysis employed in the TDF procedure must be such that compounds such as FOS and other NDO are not hydrolysed. This is demonstrated with the current procedure from the results shown in Fig. 9 and Table 1. There is no hydrolysis of most NDO, and just limited hydrolysis of Polydextrose® and Fibersol 2®, as would be expected from the known composition of these products. There is also no hydrolysis of sucrose, which would allow separate measurement of this disaccharide if required.

Table 5. Values determined for D-glucose, D-fructose and available carbohydrates in a range of samples using the current available carbohydrates procedure

Sample	D-glucose (% dwb)	D-Fructose (% dwb)	Available carbohydrates (%, dwb)[a)]
Hylon VII®	49.7	0.3	50.0
ActiStar®	62.1	0.0	62.1
FiberSym®	38.9	0.2	39.1
Fiber Rite®	80.2	0.2	80.4
Wheat starch	106.7	0.7	107.4
Regular Maize Starch	104.0	0.4	104.4
Kidney beans	46.9	0.5	47.4
Cooked potato	83.2	2.5	85.7
Cabbage	12.7	17.3	30.0
Raw swede	37.6	26.5	64.1
Carrot	35.3	26.1	61.4
Cooked cauliflower	24.0	18.3	42.3
Semi-ripe banana	37.4	22.7	60.1
Heinz® baked beans	44.6	6.2	50.8
Butter beans	46.3	1.4	47.7
Kellogg corn flakes breakfast cereal	84.7	3.2	87.9
Ryvita® Biscuits	66.7	0.6	67.3
White bread	77.8	0.2	78.0
Weetabix® breakfast cereal	74.4	2.0	76.4
Wholewheat pasta	74.9	0.4	75.3
Quick oats breakfast porridge	72.9	1.0	73.9
Kellogg Sugar Frosties® breakfast cereal	68.3	17.8	86.1
Odlum's cream plain flour	78.1	1.4	79.5
Uncle Ben's Express boiled rice	89.2	0.5	89.7
Long grain rice	94.5	0.2	94.7

a) Available carbohydrates calculated by adding D-fructose plus D-glucose.

4 Conclusions

In this paper we describe an improved method for the measurement of total dietary fibre consistent with the Codex Alimentarius definition of fibre. Several problems/challenges that became evident with our initial procedure (AOAC Method 2009.01) have been resolved; enzyme levels have been optimised allowing an incubation time (4 h) consistent with human ileostomy transit time; problems associated with measurement of FOS have been resolved by using a different HPLC system, and this was facilitated by removing glycerol from all enzymes employed in the assay; meaningful analytical results have been obtained for phosphate-crosslinked starch by modifying the incubation time and adjusting the levels of PAA and AMG employed; also, optimisation of the levels of PAA and AMG employed in the assay prevents the formation of specific resistant, branched maltodextrins (6^3,6^5-di-α-D-glucosyl maltopentaose and 6^3-α-D-glucosyl maltotriose), which in the former procedure had to be removed through an extra enzymic incubation. Most importantly, the enzymes employed to hydrolyse non-resistant starch are sufficiently pure to allow their use at quite high concentrations with no hydrolysis of fibre components and particularly of NDO. The procedure also facilitates the measurement of available carbohydrates.

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There are no conflict of interests associated with this article.

5 References

[1] Asp, N.-G. In: McCleary, B. V., Prosky, L. (Eds.), *Advanced Dietary Fibre Technology*, Blackwell Science Publishers, Oxford 2001, pp. 77–86.

[2] Hipsley, E. Dietary "fibre" and pregnancy toxaemia. *Br. Med. J.* 1953, *2*, 420–422.

[3] Trowell, H. C., Southgate, D. A. T., Wolever, T. M. S., Leeds, A. R., et al. Dietary fibre redefined. *Lancet* 1976, *1*, 967.

[4] Prosky., L, Asp, N.-G., Furda, I., DeVries, J. W., et al. Determination of total dietary fiber in foods and food products: Collaborative study. *J. AOAC Int.* 1985, *68*, 677–679.

[5] Prosky, L., Schweizer, T. F., et al. Determination of soluble dietary fiber in foods and food products: Collaborative study. *J. AOAC Int.* 1994, *77*, 690–694.

[6] Lee, S. C, Prosky, L., DeVries, J. W., Determination of total, soluble and insoluble dietary fiber in foods-enzymatic-

© 2015 The Authors Starch Published by Wiley-VCH Verlag GmbH & Co. KGaA

gravimetric method, MES-tris buffer: Collaborative study. *J. AOAC Int.* 1992, *75*, 395–416.

[7] Official Methods of Analysis of AOAC International, Methods 925.10, 985.29, 991.42, 991.43, 993.19, 994.13, 996.01, 2001.03, 2002.01, 2002.02, 2009.01, and 2011.25, 19th Edn., AOAC International, Rockville, Maryland 2012.

[8] McCleary, B. V., Sloane, N., Draga, A., Lazewska, I., Measurement of Total Dietary Fiber Using AOAC Method 2009. 01 (AACC International Approved Method 32-45.01): Evaluation and Updates. *Cereal Chem.* 2013, *90*, 396–414.

[9] Zielinski, G., DeVries, J. W., Craig, S. A., Bridges, A. R., Dietary fiber methods in Codex Alimentarius: Current status and ongoing discussions. *Ceral Foods World* 2013, *58*, 1–5.

[10] Englyst, H. N., Kingman, S. M., Cummings, J. H., Classification and measurement of nutritionally important starch fractions. *Eur. J. Clin. Nutr.* 1992, *46*, S33–S50.

[11] Englyst, H. N., Cummings, J. H., Simplified method for the measurement of non-starch polysaccharides by gas-liquid chromatography of constituent sugars as alditol acetates. *Analyst* 1984, *109*, 937–942.

[12] Englyst, H. N., Hudson, G. J., Colorimetric method for the routine measurement of dietary fibre as non-starch polysaccharides. A comparison with gas-liquid chromatography. *Food Chem.* 1987, *24*, 63–67.

[13] Englyst, H. N., Cummings, J. H., An improved method for the measurement of dietary fiber as non-starch polysaccharides in plant foods. *J. AOAC Int.* 1988, *71*, 808–814.

[14] Englyst, HN, Quigley Hudson, MEGJ., Determination of dietary fibre as non-starch polysaccharides, with gas-liquid chromatography, high-performance liquid chromatography, or spectrophotometric measurement of component sugars. *Analyst* 1994, *119*, 1497–1509.

[15] Southgate, D. A. T. Determination of carbohydrates in foods. II. Unavailable carbohydrates. *J. Sci. Food Agric.* 1969, *20*, 326–330.

[16] Southgate, D. A. T., Hudson, G. J., Englyst, H. N., The analysis of dietary fibre, the choices for the analyst. *J. Sci. Food. Agric.* 1978, *29*, 979–988.

[17] Codex Alimentarius. FAO, Rome: Joint FAO/WHO Food Standards Programme, Secretariat of the Codex Alimentarius Commission; 2010. Guidelines on nutrition labelling CAC/GL 2-1985 as last amended 2010.

[18] Howlett, J. F., Betteridge, V. A., Champ, M, Craig, S. A. S. et al. 2010, The definition of dietary fiber—discussions at the Ninth Vahouny Fiber symposium: building scientific agreement. http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2972185/.

[19] McCleary, B. V. An integrated procedure for the measurement of total dietary fiber (including resistant starch), non-digestible oligosaccharides and available carbohydrates. *Anal. Bioanal. Chem.* 2007, *389*, 291–308.

[20] McCleary, B. V., DeVries, J. W., Rader, J. I., Cohen, G., et al. Determination of total dietary fiber (CODEX definition) by enzymatic-gravimetric method and liquid chromatography: Collaborative study. *J. AOAC Int.* 2010, *93*, 221–233.

[21] McCleary, B. V., DeVries, J. W., Rader, J. I., Cohen, G., et al. Determination of insoluble, soluble, and total dietary fiber (CODEX definition) by enzymatic-gravimetric method and liquid chromatography: collaborative study. *J. AOAC Int.* 2012, *95*, 824–844.

[22] McCleary, B. V., Monaghan, D. A., Measurement of resistant starch. *J. AOAC Int.* 2002, *85*, 665–675.

[23] McCleary, B. V., McNally, M., Rossiter, P., Measurement of resistant starch by enzymic digestion in starch samples and selected plant materials: Collaborative Study. *J. AOAC Int.* 2002, *85*, 1103–1111.

[24] Read, N.W, Cammack, J., Edwards, C., Holgate, A. M., et al. Is the transit time of a meal through the small intestine related to the rate at which it leaves the stomach? *Gut* 1982, *23*, 824–828.

[25] Lönnerblad, L. Transit time through small intestine: Roentgenologic study on normal variability. *Acta Radiol. Suppl.* 1951, *88*, 1–85.

[26] Miller, M. A., Parkman, H. P., Urbain, J. L., Brown, K. L., et al. Comparison of scintigraphy and lactulose breath hydrogen test for assessment of orocecal transit: lactulose accelerates small bowel transit. *Dig. Diseases Sci.* 1997, *42*, 10–18.

[27] Geboes, K. P., Luypaerts., A., Rutgeerts, P., Verbeke, K., Inulin is an ideal substrate for a hydrogen breath test to measure the orocaecal transit time. *Aliment Pharmacol. Ther.* 2003, *18*, 721–729.

[28] Geypens, B., Bennink, R., Peeters, M., Evenepole, P., et al. Validation of the lactose-[^{13}C]ureide breath test for determination of orocecal transit time by scintigraphy. *J. Nucl. Med.* 1999, *40*, 1451–1455.

[29] Sadik, R., Abrahamsson, H., Bjornsson, E., Gunnarsdottir, A, Stotzer, P.-O., Etiology of portal hypertension may influence gastrointestinal transit. *Scand. J. Gastroenterol.* 2003, *38*, 1039–1044.

[30] Stotzer, P. O., Abrahamsson, H, Human postprandial gastric emptying of indigestible solids can occur unrelated to antral phase III. *Neurogastroenterol. Motil.* 2000, *12*, 415–419.

[31] Deiteren, A., Camilleri, M., Bharucha, A. E., Burton, D., et al. Performance characteristics of scintigraphic colon transit measurement in health and irritable bowel syndrome and relationship to bowel functions. 22, 415. *Neurogastroenterol. Motil.* 2010, *423*, e95.

[32] Zarate, N., Mohammed, S. D., O'Shaughnessy, E., Newell, M., et al. Accurate localization of a fall in pH within the ileocecal region: validation using a dual-scintigraphic technique. *Am. J. Physiol.* 2010, *299*, G1276–G1286.

[33] Camilleri, M., Thorne, N. K., Ringel, Y., Hasler, W. L., et al. Wireless pH-motility capsule for colonic transit: prospective comparison with radiopaque markers in chronic constipation. *Neurogastroenterol. Motil.* 2010, *22*, 874–882. e233.

[34] Okuma, K., Gordon, D. T., Determination of total dietary fiber in selected foods containing resistant maltodextrins by enzymatic-gravimetric method and liquid chromatography: Collaborative study (2002). *J. AOAC Int.* 2002, *85*, 435–444.

[35] Brunt, K., Sanders, P., Improvement of AOAC 2009. 01 total dietary fibre method for bread and other high starch containing matrices. *Food Chem.* 2013, *140*, 547–580.

[36] Park, W., Department of Biochemistry and Biophysics, Texas A & M University, College Station, Texas, USA, personal communication.

[37] McCleary, B. V. Modification to AOAC Official Methods 2009.01 and 2011.25 to allow for minor overestimation of low molecular weight soluble dietary fiber in samples containing starch. Study. *J. AOAC Int.* 2014, *97*, 896–901.

[38] Dahlqvist, A., Assay of intestinal disaccharidases. *Anal. Biochem.* 1968, *22*, 99–107.

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[39] Maningat, C. C, Seib, P. A., Bassi, S. D., Dietary fiber content of cross-linked phosphorylated resistant starch (RS4) determined by the Prosky and McCleary methods. Part I. Factors affecting in vitro digestion of starch in a food sample. *Cereal Food World* 2013, *58*, 247–251.

[40] Champ, M., Martin, L., Noah, L., Gratas, M, In: Cho, S. S., Prosky, L., Dreher, M. (Eds.) *Complex Carbohydrates in Foods*, Marcel Dekker, Inc, New York 1999, pp. 169–187.

[41] Champ, M., Determination of resistant starch in foods and food products: Interlaboratory study. *Eur. J. Clin. Nutr.* 1992, *46*, S51–S62.

[42] Muir, J. G., O'Dea, K., Measurement of resistant starch: factors affecting the amount of starch escaping digestion in vitro. *Am. J. Clin. Nutr.* 1992, *56*, 123–127.

[43] Faisant, N., Planchot, V., Kozlowski, F., Pacouret, M.-P., et al. Resistant starch determination adapted to products containing high levels of resistant starch. *Sci. Aliments* 1995, *15*, 83–89.

[44] Goni, I., Garcia-Diz, E., Manas, E., Saura-Calixto, F., Analysis of resistant starch: A method for food and food products. *Food Chem.* 1996, *56*, 445–449.

[45] Akerberg, A. K. E., Liljberg, G. M., Granfeldt, Y. E., Drews, A. W., Bjorck, M. E., An in vitro method based on chewing, to predict resistant starch content in foods allows parallel determination of potentially available starch and dietary fiber. *J. Nutr.* 1998, *128*, 651–660.

[46] Champ, M., Kozlowski, F., Lecannu, G., In: McCleary, B. V., Prosky, L. (Eds.), *Advanced Dietary Fibre Technology*, Blackwell Science Ltd, Oxford 2000, pp. 106–119.

[47] Tanabe, K., Nakamura, S., Oku, T., Inaccuracy of AOAC Method 2009. 01 with Amyloglucosidase for measuring non-digestible oligosaccharides and proposal for an improvement of the method. *Food Chem.* 2014, *151*, 539–546.

[48] Shi, Y-S., Jeffcoat, R., In: McCleary, B. V., Prosky, L. (Eds.), *Advanced Dietary Fibre Technology*, Blackwell Science Ltd, Oxford 2000, pp. 430–439.

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